Statistical Analysis of Networks Data



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Chapter 1

Network Models and Exchangeability

These notes concern the the stochastic properties of networks data. In this context, a network represents interactions between entities (nodes) where the presence of an interaction is indicated by an edge. Hence, a network is a graph $G = (V, E)$ where $E \subseteq [V]^2$.
We will say that a vertex v is incident with an edge e if $v \in e$, and that two vertices are adjacent/neighbours if they are connected by an edge.

1.1 Basics in Graph Theory

In this section we recall some basic definitions and properties in Graph Theory.

Definition 1.1.1. *A network is described by its **adjacency matrix** A. If node i and j are connected then $A_{ij} = 1$, if not $A_{ij} = 0$.*

Remark 1.1.2. *A network can also be represented by a list of edges, an edge list, that specifies the existing edges $\{(i,j)\}_{i,j}$.*

Definition 1.1.3. *A graph $H = (V(H), E(H))$ is a **subgraph** of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.*

Definition 1.1.4. *The **incidence matrix** C catalogues the relationship between nodes and edges. If vertex v_i and edge e_j are incident then $C_{ij} = 1$, if not $C_{ij} = 0$.*

Definition 1.1.5. *The **degree vector** is defined as:*

$$d_i = \sum_{j \neq i} A_{ij}. \tag{1.1}$$

Definition 1.1.6. *The **complete graph** of n nodes is written as K_n.*

Definition 1.1.7. *A **cycle** on n nodes is written as C_n. Every node in a cycle has degree 2.*

Definition 1.1.8. *A* **d-*regular*** *graph is a graph where all nodes have the same degree d.*

Definition 1.1.9. *A connected graph with no cycles is a* ***tree****. A disjoint union of trees is a* ***Forest****.*

Definition 1.1.10. *A* ***bipartite graph*** $G = (V(G), E(G))$ *is one where the vertices can be split into* $V_1(G)$ *and* $V_2(G)$*, and where an edge has one endpoint in* $V_1(G)$ *and the other in* $V_2(G)$*.*

Definition 1.1.11. *A* ***path/walk*** *on graph G from vertex* v_0 *to* v_i *is an alternating sequence* $\{v_0, e_1, v_1, e_2, ..., v_i\}$*.*

Definition 1.1.12. *A graph is* ***connected*** *if it is not empty and any two vertices in the network are connected by a path/walk.*

Definition 1.1.13. *If we equip edges with weights then we say that we* ***decorate*** *the graph with auxiliary numerical values.*

Proposition 1.1.14. *The total number of edges in a network is:* $\sum_i d_i = 2|E|$*.*

1.2 Network Models

In this section we present several common network models.

1.2.1 The Erdős-Rényi Network

An Erdős-Rényi network on n nodes with edge probability $0 < p < 1$, $\mathrm{ER}(n,p)$, is one of the simplest graph model. The adjacency matrix is generated element by element as:

$$A_{ij} \sim \mathrm{Ber}(p), \quad 1 \leq j < i \leq n. \tag{1.2}$$

where each realisation is independent.

Proposition 1.2.1. *In* $\mathrm{ER}(n,p)$*:* $d_i \sim \mathrm{Bin}(n-1, p)$*.*

1.2.2 The Chung-Lu/Configuration Model

The simplest generalisation of the previous model is the Chung-Lu/Configuration Model which introduces n parameters π_i and then generates edges independently by:

$$A_{ij} \sim \mathrm{Ber}(\min(\pi_i \pi_j, 1)), \quad 1 \leq j < i \leq n. \tag{1.3}$$

A common choice for π_i is to use some power laws, e.g. $\pi_i = \theta i^{-\gamma}$.

1.2.3 The Random Geometric Graph

A Random Geometric Graph is a network that has a strong latent spatial structure. Such a graph is generated by randomly placing n locations x_i in a specified spatial domain, specifying a radius $r > 0$ and connecting node i and j. Hence, $A_{ij} = 1$ if $d(x_i, x_j) < r$ for some specified distance $d(x, y)$.

1.2.4 The Inhomogeneous Random Graph

In an Inhomogeneous Random Graph, edges are generated independently by:

$$A_{ij} \sim \text{Ber}(p_{ij}), \quad 1 \leq j < i \leq n. \tag{1.4}$$

1.2.5 The Stochastic Block Model

The Stochastic Block Model is a graph that clearly displays group structure. For each node i we define a random variable z_i that takes the value $\{1, ..., k\}$, where this variable is indicating the group membership of node i. We additionally define a connection probability matrix Θ which has entries θ_{ab} for $1 \leq a \leq b \leq k$. Then, conditionally on A_{ij}, we have:

$$A_{ij}|\{z_i, z_j\} \sim \text{Ber}(\theta_{z_i z_j}), \quad 1 \leq j < i \leq n. \tag{1.5}$$

where each realisation is independent.

1.2.6 The Mixed Membership Model

A generalisation of the Stochastic Block Model is the Mixed Membership Model. Generate latent variables ξ_i for each node i from the Dirichlet distribution of dimension k with parameters α. Define $\Theta = (\theta)_{pq}$ and we draw:

$$A_{ij}|\{\xi_i, \xi_j\} \sim \text{Ber}(\xi_i^T \Theta \xi_j). \tag{1.6}$$

1.2.7 The Random Dot Product Graph

The Random Dot Product Graph (RDPG) is a latent position model of:

$$\mathbb{E}[A_{ij}|\Xi] = \rho_n \cdot \xi_i^T \xi_j, \tag{1.7}$$

where the latent positions ξ_i of node i are generated i.i.d. by some probability density $f(\xi)$.

Proposition 1.2.2. *In the RDPG:* $\mathbb{E}[d_i] = \rho_n \sum_{j \neq i} \mathbb{E}\left[\xi_i^T \xi_j\right] = \rho_n \sum_{j \neq i} \mathbb{E}[\xi_i^T] \mathbb{E}[\xi_j]$.

Proof. We use the Tower Property to find:

$$\mathbb{E}[d_i] := \mathbb{E}\left[\sum_{j \neq i} A_{ij}\right] = \sum_{j \neq i} \mathbb{E}[A_{ij}] = \sum_{j \neq i} \mathbb{E}\left[\mathbb{E}[A_{ij}|\Xi]\right] = \sum_{j \neq i} \mathbb{E}\left[\rho_n \cdot \xi_i^T \xi_j\right] = \rho_n \sum_{j \neq i} \mathbb{E}\left[\xi_i^T \xi_j\right]. \tag{1.8}$$

□

1.2.8 The Degree Corrected Stochastic Block Model

The Degree Corrected Stochastic Block Model is a model where for each node i we define a random variable z_i that takes its values in $\{1,...,k\}$, where this variable is indicating the group membership of node i, and a latent uniform ξ_i. We define a connection probability matrix Θ which has entries θ_{ab} for $1 \leq a \leq b \leq k$. Then with a 1-dimensional function $g(x)$ we draw:

$$A_{ij}|\{z_i, z_j, \xi_i, \xi_j\} \sim \text{Ber}(\theta_{z_i z_j} + g(\xi_i)g(\xi_j)), \quad 1 \leq j < i, \tag{1.9}$$

where each realisation is independent.

1.3 Exchangeability

Most models fall in a more general framework of permutation invariance. Namely, that for most of the enumerated model the value of i or j contain no information about the model structure. Thus if we introduce a permutation that remaps all indices, the nature of the model will not change.
In this section we consider a permutation on the ordering, Π, so that $\Pi(\{1,...,n\}) = (\pi(1),...,\pi(n))$. Furthermore, for an adjacency matrix A, we consider A^Π to be the permuted adjacency matrix.

Definition 1.3.1. ***Permutation-invariance*** *of the distribution holds when* $\mathbb{P}(A = a) = \mathbb{P}(A^\Pi = a)$ *for any permutation* Π *and any adjacency matrix* A*. That is, permuting the adjacency matrix does not change its distribution.*

Definition 1.3.2. *We denote by* $\mathbf{Sym}(\mathbf{A})$ *the group of all permutations of a set* A*.*

Definition 1.3.3. *Let* E *be a suitable space. A sequence of* E*-valued random variables* $(X_n)_{n\in\mathbb{N}}$ *is* ***exchangeable*** *if, for all* $\Pi \in Sym(\mathbb{N})$*:*

$$(X_n)_n = \left(X_{\Pi(n)}\right)_n \quad \textit{in distribution.} \tag{1.10}$$

Definition 1.3.4. *For any* $k \geq 1$ *we consider an array of* E*-valued random variables* $(X_e)_{e\in\mathbb{N}^{(k)}}$ *indexed by size-*k *subsets of* $\mathbb{N}$*, and say it is* ***(jointly) exchangeable*** *if, for all* $\Pi \in Sym(\mathbb{N})$*:*

$$(X_e)_e = \left(X_{\Pi(e)}\right)_e \quad \textit{in distribution,} \tag{1.11}$$

where if $e = \{n_1,...,n_k\}$ *then* $\Pi(e) := \{\Pi(n_1),...,\Pi(n_k)\}$*.*

General arrays were studied by Hoover, Aldous, Fremlin, Talagrand and Kallenberg. Finite exchangeability simply puts e in a finite space. A finite $n \times m$ random matrix A is row-column exchangeable if for a n-permutation σ, a m-permutation π and for all Borel sets $N_{11},...,N_{nm}$:

$$\mathbb{P}(A_{11} \in N_{11}, A_{12} \in N_{12},..., A_{nm} \in N_{nm}) = \mathbb{P}(A_{\sigma(1)\pi(1)} \in N_{11}, A_{\sigma(1)\pi(2)} \in N_{12},..., A_{\sigma(n)\pi(m)} \in N_{nm}).$$

Definition 1.3.5. *For $r > n$ and $q > m$ the matrix A is called* $(\mathbf{r},\mathbf{q})$***-extendible*** *if there are matrices T,Z and W with:*

$$T = (A_{ij})_{i,j}, \quad i = \{1,...,n\}, \quad j = \{m+1,...,q\}, \tag{1.12}$$

$$Z = (A_{ij})_{i,j}, \quad i = \{n+1,...,r\}, \quad j = \{1,...,m\}, \tag{1.13}$$

$$W = (A_{ij})_{i,j}, \quad i = \{n+1,...,r\}, \quad j = \{m+1,...,q\}, \tag{1.14}$$

such that:

$$A^* = \begin{pmatrix} A & T \\ Z & W \end{pmatrix}, \tag{1.15}$$

is row-column exchangeable. A matrix that is (r,q)-extendible for all $r > n$ and $q > m$ is called ***infinitely extensible****.*

Theorem 1.3.6 (Aldous Hoover)**.** *An array A is jointly exchangeable if and only if it has the same distribution as:*

$$A_{ij} = f(\alpha,\xi_i,\xi_j,\zeta_{ij}), \quad 1 \leq i < j, \tag{1.16}$$

with $f : \mathbb{R}^4 \longrightarrow \mathbb{R}$ and some i.i.d. uniform random variables α, ξ_i and ζ_{ij}.

It is usually convenient to construct models conditional on latent random variables since the latent variables are all i.i.d. and hence the distribution is the same whichever ordering we pick. Therefore, the permuted random variables have the same distribution as the unpermuted ones. However, the realisations are not the same unpermuted and permuted. Finally, another advantage of considering such models is that some statistics on them are also permutation invariant.

Example 1.3.7. *Assume that $\xi_i \sim \text{Unif}(0,1)$ i.i.d. such that:*

$$A_{ij}|\xi \sim \text{Ber}(f(\xi_i,\xi_j)).$$

Now if we apply a permutation to the order we obtain that $\xi_{\pi(i)} \sim \text{Unif}(0,1)$ i.i.d., and so:

$$A_{\pi(i)\pi(j)}|\xi \sim \text{Ber}(f(\xi_{\pi(i)},\xi_{\pi(j)})),$$

has the same distribution.

Chapter 2

Statistics for Networks

Summary statistics are used to summarize a set of observations, in order to communicate the largest amount of information as simply as possible. To define statistical summaries for networks we need to understand what natural network summaries are.
A basic concept is to count the occurence of subgraphs in G.

Definition 2.0.1. *A **subgraph count** simply corresponds to the number of copies of F in G:*

$$X_F(G) := \sum_{F' \subset G} \mathbb{1}_{\{F' \cong F\}}. \tag{2.1}$$

Here, $\cong$ means isomorphic, implying that F' can be mapped to F.

One problem with $X_F(G)$ is that it is not invariant to the number of nodes n, or the marginal probability of getting an edge $\rho = \mathbb{P}(A_{ij} = 1) > 0$.

Example 2.0.2. *Suppose A is generated as an Erdős-Rényi graph with marginal edge probability ρ, then with K_n the complete graph on n nodes we have:*

$$\begin{aligned}
\mathbb{E}_G[X_F(G)] = \mathbb{E}_G\left[\sum_{F' \subset G} \mathbb{1}_{\{F' \cong F\}}\right] &= \mathbb{E}_G\left[\sum_{F' \subset K_n} \mathbb{1}_{\{F' \cong F\}} \mathbb{1}_{\{F' \subseteq G\}}\right] \\
&= \sum_{F' \subset K_n} \mathbb{1}_{\{F' \cong F\}} \mathbb{E}_G[\mathbb{1}_{\{F' \subseteq G\}}] \\
&= \sum_{F' \subset K_n} \mathbb{1}_{\{F' \cong F\}} \mathbb{P}(F' \subseteq G) \\
&= \mathbb{P}(F \subseteq G) \sum_{F' \subset K_n} \mathbb{1}_{\{F' \cong F\}} \\
&= \rho^{|E(F)|} \sum_{F' \subset K_n} \mathbb{1}_{\{F' \cong F\}} \\
&= \rho^{|E(F)|} X_F(K_n).
\end{aligned}$$

This example tells us how $X_F(G)$ scales with ρ. In addition, $X_F(K_n)$ is deterministic and takes the form of $(n)_{|V(F)|}/\mathrm{aut}(F)$ where the notation $(n)_k$ is defined as $(x)_y = x(x-1)...(x-y+1)$.

Definition 2.0.3. *We define* **aut(G)** *as:*

$$|\{\phi \in Sym(v(G)) : pq \in E(G) \iff \phi(p)\phi(q) \in E(G)\}|. \tag{2.2}$$

It is the order of the automorphism group of G, i.e. the number of adjacency-preserving permutations of $V(G)$.

Example 2.0.4. *The size of the automorphism group of an edge is 2, since permuting two labels will leave the adjacency matrix invariant.*

Example 2.0.5. *The size of the automorphism group of a triangle is $3! = 6$, since we first choose one out of the three nodes, then choose one out of the two remaining vertices.*

Example 2.0.6. *The automorphism group of the complete graph K_n is S_n which is the group of all permutations of $\{1, ..., n\}$. The size of the group is $n!$.*

Definition 2.0.7. *We define the* ***average degree of a graph*** *on n nodes with e edges as:*

$$\overline{d} := \frac{2e}{n}. \tag{2.3}$$

We furthermore define the ***maximum average degree of a graph*** *F as:*

$$m(F) := \max_{H \subset F} \{\overline{d}(H)\}. \tag{2.4}$$

Definition 2.0.8. *A graph H is said to be* ***balanced*** *if $m(H) = d(H)$ and is* ***strictly balanced*** *if $F \subseteq H$ and $\overline{d}(F) = m(H)$ implies that $F = H$.*

Example 2.0.9. • *Trees, cycles and complete graphs are strictly balanced.*
• *It has been shown that $X_F(G)$ has an asymptotically Gaussian distribution as long as F is a strictly balanced graph.*

Theorem 2.0.10 (Poisson Limit)**.** *Suppose that H is a strictly balanced graph with k vertices and $l \geq 2$ edges, and its automorphism group has μ_a members. Let $c > 0$ be a constant and set $p := cn^{-k/l}$. For G generated as an Erdős-Rényi graph with success probability p denote by $X_H(G)$ the number of copies of H in G. Then $X_H(G)$ asymptotically becomes a Poisson random variable with mean c^l/μ_a.*

Remark 2.0.11. *The edge probability p has been specified using:*

$$\mathbb{E}[X_F(G)] = \frac{(cn^{-k/l})^l (n)_k}{\mu_a} \approx \frac{c^l}{\mu_a}. \tag{2.5}$$

If we define G_n as the Stochastic Block Model on Θ with K groups with h_b nodes in group b. We furthermore define a vector $\underline{c}$, where c_i indicates the group of node $1 \leq i \leq n$. We now count a strictly balanced graph H in G_n. We define:

$$\lambda := \mathbb{E}[X_H(G_n)] = \binom{n}{|V(H)|} \frac{(|V(H)|)!}{aut(H)} \mu(H), \tag{2.6}$$

where for $\mathcal{I} = \{1 \leq u < v \leq |V(H)| : (u,v) \in E(H)\}$:

$$\mu(H) = \sum_{c_1,...,c_{V(H)}}^{K} h_{c_1}...h_{c_{|V(H)|}} \prod_{u,v \in \mathcal{I}} \theta_{c_u c_v}. \tag{2.7}$$

It has been shown by Coulson et al. (2015) that the distribution of $X_H(G_n)$ becomes Poisson with mean λ. Note that the label vector is random, but that the proportions put in each group converges to a fixed proportion. Therefore $\mu(H)$ is deterministic.

2.1 Network Descriptors

In this section, we consider G to be an unlabelled graph and $\{v_1,...,v_k\}$ a sequence of vertices in G. We define three types of random graphs:

- $G(v_1,...,v_k)$ is obtained by randomly sampling the vertices uniformly, where we allow $v_i = v_j$ but no self-loops.
- Let $G[k]$, for $k \geq 1$, be the random graph obtained by sampling $\{v_1,...,v_k\}$ uniformly at random among the vertices of G, with replacement.
- For $k \leq V(G)$, we let $G[k]'$ to be the random graph $G(v'_1,...,v'_k)$ where we sample $\{v'_1,...,v'_k\}$ uniformly at random without replacement.

Definition 2.1.1. *The functional* $\mathbf{t}(\mathbf{F},\mathbf{G})$ *is defined for two graphs* F *and* G *as the proportion of all mappings* $V(F) \longrightarrow V(G)$ *that are graph homomorphisms* $F \longrightarrow G$*, i.e. it maps adjacent vertices to adjacent vertices.*
In probabilistic terms, $t(F,G)$ *is the probability that a uniform random mapping* $V(F) \longrightarrow V(G)$ *is a graph homomorphism. Assuming* F *is labelled and* $k = |V(F)|$ *we define:*

$$t(F,G) := \mathbb{P}(F \subseteq G[k]). \tag{2.8}$$

We note that both F and $G[k]$ are graphs on $[k]$, so the relation $F \subseteq G[k]$ is well-defined as containment of labelled graphs on the same vertex set. Furthermore, although the relation $F \subseteq G[k]$ may depend on the labelling of F, the probability $t(F,G)$ does not, by symmetry, hence $t(F,G)$ is well-defined for unlabelled F and G.
Now if we define $t_{inj}(F,G) := \mathbb{P}(F \subseteq G[k]')$ and $t_{ind}(F,G) := \mathbb{P}(F = G[k]')$ for $|V(F)| \leq |V(G)|$ (whilst if $|V(F)| > |V(G)|$ then $t(F,G) = 0$), and since the probability that a random sample $\{v_1,...,v_k\}$ of vertices in G contains some repeated vertex is $\leq k^2/(2|V(G)|)$ it follows that:

$$|t(F,G) - t_{inj}(F,G)| \leq \frac{|V(F)|^2}{2|V(G)|}. \tag{2.9}$$

And in fact, we can define a metric between two networks G_1 and G_2 by computing for an infinite

sequence of subgraphs $\{F_j\}_j$:

$$d(G_1, G_2) := \sum_i 2^{-i} |t(F_i, G_1) - t(F_i, G_2)|. \tag{2.10}$$

This metric can be shown to be equivalent to the cut metric introduced by Lovasz et al.

Definition 2.1.2. *We denote by* $\mathbf{P_k^{(n)}}$ *the proportion of vertices with degree* k *in* G*, where* $|V(G)| = n$*. Then we may write the empirical summary as:*

$$P_k^{(n)} := \frac{1}{n} \sum_i \mathbb{1}_{\{d_i = k\}}. \tag{2.11}$$

Sometimes $\{P_k^{(n)}\}$ *is referred to as the degree distribution of* G*.*

Of course the expectation of $P_k^{(n)}$ is given by:

$$\mathscr{P}_k^{(n)} := \mathbb{E}[P_k^{(n)}] = \frac{1}{n} \sum_i \mathbb{P}(d_i = k). \tag{2.12}$$

And if we can assume that $\mathscr{P}_k^{(n)} \longrightarrow p_k$ for some deterministic constants $\{p_k\}$ then we can characterize the degree distribution of G.

Definition 2.1.3. *A random graph* G *is said to be* ***scale-free*** *with exponent* τ *if the following limit exists:*

$$\lim_{k \to \infty} \frac{\log(p_k)}{\log(1/k)} = \tau. \tag{2.13}$$

2.2 Fitting Network Models

- Recall that for an Erdős-Rényi network we need to estimate the following estimated edge density:

$$\widehat{\rho} = \frac{\sum_{i<j} A_{ij}}{\binom{n}{2}}. \tag{2.14}$$

Now we can calculate the moments of this estimator, where from the independence of the trials:

$$\mathbb{E}(\widehat{\rho}) = \frac{\sum_{i<j} \rho}{\binom{n}{2}} = \rho, \tag{2.15}$$

$$\mathrm{Var}(\widehat{\rho}) = \frac{\sum_{i<j} \rho(1-\rho)}{\binom{n}{2}^2} = \frac{\rho(1-\rho)}{\binom{n}{2}}. \tag{2.16}$$

Hence using standard Central Limit Theorems we can deduce that a function of $\sum_{i<j} A_{ij}$ becomes Gaussian if ρ is sufficiently large. And we have:

$$\frac{1}{\sqrt{\binom{n}{2}\rho(1-\rho)}} \left(\sum_{i<j} A_{ij} - \binom{n}{2} \rho \right) \xrightarrow{d} \mathcal{N}(0,1). \tag{2.17}$$

- Considering another example, for the Chung-Lu model we need to estimate:

$$\widehat{\pi}_i := \frac{d_i}{\sqrt{\|d\|_1}} = \frac{d_i}{\sqrt{\sum_i d_i}}, \quad i = 1, ..., n. \tag{2.18}$$

In fact, with this model, we can determine that d_i becomes Gaussian under suitable conditions. As the denominator converges in distribution to a non-zero constant, this leads us to keep the Gaussian distribution for the estimator. This is not a standard setup since the number of parameters (i.e. dimensionality of π) increases with n.
We can first start by understanding d_i better. Each network degree is a sum of $(n-1)$ Bernoulli $\text{Ber}(\pi_i\pi_j)$ random variables. Now the probability that d_i takes the value k is the sum of all distinct ways in which k successes can occur in $(n-1)$ $\text{Ber}(\pi_i\pi_j)$ trials. Hence this yields a Poisson-Binomial distribution, which is a generalization of the Binomial distribution by summing Bernoullis with different success probabilities.
The key to understanding this distribution are to find the moments. We can note that if π is a deterministic vector of parameters, and let d_i be the degree vector of an n-node simple graph whose edges are independent $\text{Ber}(\pi_i\pi_j)$ trials then:

$$\mathbb{E}(d_i) = \pi_i(\|\pi\|_1 - \pi_i), \tag{2.19}$$

$$\text{Var}(d_i) = \pi_i(\|\pi\|_1 - \pi_i) - \pi^2 \sum_{j \neq i} \pi_j^2, \tag{2.20}$$

$$\text{Cov}(d_i, d_j) = \pi_i\pi_j(1 - \pi_i\pi_j). \tag{2.21}$$

Another key characterizer of the distribution of d_i is its dispersion. In the case of the Chung-Lu model it is given by:

$$\frac{\text{Var}(d_i)}{\mathbb{E}(d_i)} = 1 - \pi_i \frac{\sum_{j \neq i} \pi_j^2}{(\|\pi\|_1 - \pi_i)}. \tag{2.22}$$

2.3 Non-Parametric Summaries

In this section we will describe some tools to know how central or important some vertices are. Degrees are of course one way of determining how important a particular node i is.
First define $\text{dist}_G(u, v)$ as the graph distance between nodes u and v in G, i.e. the minimal number of edges in any path linking u and v. Note that if no path exists then we set the distance as ∞.

Definition 2.3.1. *We define the **closeness centrality** of vertex i in a connected graph G with n nodes as:*

$$C_i := \frac{n}{\sum_{j \neq i} dist_G(i, j)}. \tag{2.23}$$

The closeness centrality plays an important role in understanding the functionality of a network, for example when information or disease is spreading on it.

Remark 2.3.2. *Usually, to understand the full network, C_i is only computed over nodes in connected components. Otherwise, if a graph has one disconnected component, the closeness centrality of all nodes will be 0. If this is the case, one computes C_i over connected subgraphs.*

Definition 2.3.3. *We define the* ***harmonic centrality*** *of vertex i in a graph G with n nodes as:*

$$C_i^{(H)} := \sum_{j \neq i} \frac{1}{\text{dist}_G(i,j)}. \tag{2.24}$$

Definition 2.3.4. *We define the* ***average efficiency*** *of a graph G with n nodes as:*

$$E(G) := \frac{1}{n(n-1)} \sum_{i \neq j} \frac{1}{\text{dist}_G(i,j)}. \tag{2.25}$$

Definition 2.3.5. *We define the* ***local efficiency*** *of vertex i in a graph G with n nodes, with G_i as the neighbours of i, as:*

$$E := \frac{1}{n} \sum_{i \in V(G)} E(G_i). \tag{2.26}$$

Note that this measure naturally takes account of the fact that some nodes are in different connected components.

Definition 2.3.6. *We define the* ***betweenness centrality*** *of vertex i in a graph G with n nodes, with n_{jk}^i as the number of shortest paths from j to k that pass through i and with n_{jk} as the number of shortest paths between j and k, as:*

$$B_i := \sum_{1 \leq j < k \leq n} \frac{n_{jk}^i}{n_{jk}}. \tag{2.27}$$

Definition 2.3.7. *Let u be the eigenvector with eigenvalue λ of the largest eigenvalue (with positive entries due to the Perron-Frobenius theorem) of the adjacency matrix. We define the* ***eigenvector centrality*** *of vertex i in a graph G with n nodes as:*

$$C_i := \frac{1}{\lambda} \sum_{ij \in E(G)} u_j. \tag{2.28}$$

Now, for the whole network, we are interested in its degree of clustering. For a graph on the nodes $\{1, ..., n\}$ we let the number of paths with 3 nodes to be:

$$X_{P_3}(G) := \frac{1}{2} \sum_{1 \leq i,j,k \leq n} \mathbb{1}_{\{ij,\ jk \in E(G)\}}. \tag{2.29}$$

Definition 2.3.8. *We define the* ***clustering coefficient*** *of a graph G with n nodes as:*

$$CC_G := \frac{X_{C_3}(G)}{X_{P_3}(G)}, \tag{2.30}$$

where X_{C_3} is the number of cycles of 3 nodes in the graph, i.e. the number of triangles.

Definition 2.3.9. *A sequence of graphs $\{G_n\}_n$ is said to be **highly clustered** if:*

$$\liminf_{n\to\infty} CC_{G_n} > 0. \tag{2.31}$$

Lemma 2.3.10. *For any adjacency matrix A of a graph, it holds that:*

$$X_{C_3} = \frac{1}{6}\text{Trace}(A^3). \tag{2.32}$$

Lemma 2.3.11. *For any adjacency matrix A of a graph, it holds that:*

$$X_{P_3} = \frac{1}{2}\sum_{i=1}^{n} d_i(d_i - 1). \tag{2.33}$$

Chapter 3

The Stochastic Block Model (SBM)

In this chapter we will look at the SBM and how to estimate its parameters. In particular, we will see how to estimate $\{z_i\}$ and $\{\theta_{ab}\}$. Methods for doing so will follow spectral clustering and network modularity.

Block Models are usually split into types:
1) The first one is assortative stochastic blocks, meaning that the probability of connections within communities is higher than in between communities. For example, $\theta_{aa} \geq \theta_{ab}$ for $a, b \in \{1, ..., k\}$, and this shows edges inside groups.
2) The second is disassortative stochastic blocks, that is the probability of connections between communities is higher than within communities. For example, $\theta_{aa} \leq \theta_{ab}$ for $a, b \in \{1, ..., k\}$. People connect as they have different functions.
3) The third is ordered stochastic blocks that shows hierarchical structure in the graph.
4) Finally we have core-periphery structure which has a dense corse but a sparse periphery.

3.1 Maximum Likelihood Estimator (MLE)

At the basic level, the simplest SBM are $\binom{n}{2}$ independent Bernoulli trials. We can formulate the likelihood dependent on link parameters $\{\theta_{ab}\}$ and a group label $z_i \in \{1, ..., k\}$ for $i = 1, ..., n$ when there are k groups. We then find:

$$\mathbb{P}(A) = \prod_{i<j} \theta_{z_i z_j}^{A_{ij}} (1 - \theta_{z_i z_j})^{1 - A_{ij}}. \tag{3.1}$$

Therefore, the log-likelihood takes the form of:

$$l(\theta, z) = \sum_{i<j} \Big[A_{ij} \log(\theta_{z_i z_j}) + (1 - A_{ij}) \log(1 - \theta_{z_i z_j}) \Big]. \tag{3.2}$$

Now consider $h_b := \sum_i \mathbb{1}_{\{z_i = b\}}$ and $h_{ab} := \sum_{i<j} \mathbb{1}_{\{z_i = a\}} \mathbb{1}_{\{z_j = b\}}$ such that for $a \neq b$ we have $h_{ab} = h_a h_b$. Hence the log-likelihood transforms into:

$$l(\theta, z) = \sum_{a<b} \Big[h_{ab} \overline{A}_{ab}(z) \log(\theta_{ab}) + (h_{ab} - h_{ab} \overline{A}_{ab}(z)) \log(1 - \theta_{ab}) \Big], \tag{3.3}$$

where $\overline{A}_{ab}(z) := h_{ab}^{-1} \sum_{i<j} A_{ij} \mathbb{1}_{\{z_i=a\}} \mathbb{1}_{\{z_j=b\}}$. As usual, we aim to maximize the likelihood for any fixed choice of z. For the first derivative we find:

$$\frac{\partial l(\theta, z)}{\partial \theta_{ab}} = h_{ab} \frac{\overline{A}_{ab}(z)}{\theta_{ab}} - h_{ab} \frac{1 - \overline{A}_{ab}(z)}{1 - \theta_{ab}}. \tag{3.4}$$

Then setting the first derivative equal to zero we obtain:

$$\frac{\partial l(\theta, z)}{\partial \theta_{ab}} = 0 \Longrightarrow \widehat{\theta}_{ab} = \overline{A}_{ab}(z), \quad 1 \leq a \leq b \leq k. \tag{3.5}$$

Now checking that this estimator is a maximum:

$$\frac{\partial^2 l(\theta, z)}{\partial \theta_{ab}^2} = -h_{ab} \frac{\overline{A}_{ab}(z)}{\theta_{ab}^2} - h_{ab} \frac{1 - \overline{A}_{ab}(z)}{(1 - \theta_{ab})^2} < 0. \tag{3.6}$$

Finally, the profile likelihood then becomes:

$$l(z) = \sum_{i<j} \left[A_{ij} \log(\widehat{\theta}_{z_i z_j}) + (1 - A_{ij}) \log(1 - \widehat{\theta}_{z_i z_j}) \right]. \tag{3.7}$$

The final step is to maximize $l(z)$ in z to obtain an estimate of the label vector z, however this is not computationally feasible and we need to find some other ways to study clustering.

3.2 Network Modularity

Definition 3.2.1. *We define the* ***network modularity*** *for a label vector z as:*

$$\widehat{Q}_G(z) := \sum_{i<j} \left(A_{ij} - \frac{d_i d_j}{\sum_l d_l} \right) \delta_{z_i z_j}. \tag{3.8}$$

The idea with modularity is to maximise $\widehat{Q}_G(z)$ so that we arrive to a division of the nodes. This is often done hierarchically, but it comes with no statistical guarantees. Maximising modularity only works for assortative block models $\theta_{aa} > \theta_{ab}$ for all $b \neq a$. However, the likelihood estimation of z for block models works also for graphs that are not assortative. We furthermore have to decide when to stop, and we do when we have no further improvement in the modularity result. Note that whenever we need to estimate z there is a fundamental invariance as the name of the label does not matter.
This method treats the stochastic block model parametrically, where k is assumed to be known.

3.3 Spectral Clustering with the Laplacian Algorithm

This method gives us a way to cluster the points $(A_i)_i$ in $\mathbb{R}^k$. Clustering means dividing up a set of n points into $k \leq n$ groups. For each k we define a vector C_k of those group labels.
To cluster we need a measure of similarity and a distance $d(x, y)$. Similarity quantifies how similar two vectors are, like a covariance measure. On the other hand, distance quantifies how far apart two points are or how dissimilar they are. Note that clustering normally requires assumptions on assortativeness.

Hence a cluster can be described in 4 points:
1) Items that have very similar or the same properties,
2) Items whose distance is small, or their dissimilarity is small,
3) Have *contacts* with other items in the same cluster,
4) Are clearly different from the items in other clusters.

Definition 3.3.1. *We define the* ***Laplacian*** *in terms of the adjacency matrix A as:*

$$L := D - A = \text{diag}(d_1, ..., d_n) - A. \tag{3.9}$$

Theorem 3.3.2. *The matrix L satisfies the following properties:*
1) For every vector $f \in \mathbb{R}^n$ we have:

$$f^T L f = \frac{1}{2} \sum_{i,j=1}^{n} a_{ij}(f_i - f_j)^2. \tag{3.10}$$

2) L is symmetric and positive semi-definite.
3) The smallest eigenvalue of L is zero, and the corresponding eigenvector is the constant vector.
4) L has n non-negative, real-valued eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq ... \leq \lambda_n$.

Proof. 1) We see that by the definition of d_i:

$$\begin{aligned} f^T L f = f^T D f - f^T A f &= \sum_{i=1}^{n} d_i f_i^2 - \sum_{i,j} f_i f_j a_{ij} \\ &= \frac{1}{2}\left(\sum_{i=1}^{n} d_i f_i^2 - 2 \sum_{i,j=1}^{n} f_i f_j a_{ij} + \sum_{j=1}^{n} d_j f_j^2 \right) \\ &= \frac{1}{2} \sum_{i,j=1}^{n} a_{ij}(f_i - f_j)^2. \end{aligned}$$

2) The symmetry of L follows directly from the symmetry of A and D. The positive semi-definiteness is a direct consequence of 1), since we have that $f^T L f \geq 0$ for all $f \in \mathbb{R}^n$.

3) First note that $L\mathbb{1} = (D - A)\mathbb{1} = d - d = 0$. This shows that $\mathbb{1}$ is an eigenvector with eigenvalue 0.

4) For a matrix to be positive semi-definite we require $f^T L f \geq 0$. But at the same time, for an eigenvector f and using that symmetric matrices have real-valued eigenvalues we obtain:

$$f^T L f = f^T \lambda f = \lambda \|f\|^2 \geq 0. \tag{3.11}$$

Therefore, the smallest value an eigenvalue can take is 0.

□

Proposition 3.3.3 (Laplacian Algorithm)**.** *Let G be an undirected graph with non-negative weights. Then the multiplicity k of the eigenvalue 0 of L equals the number of connected components $C_1,...,C_k$ in the graph. The eigenspace of eigenvalue 0 is spanned by the indicator vectors $\mathbb{1}_{C_1},...,\mathbb{1}_{C_k}$ of those components.*
1) Input: Adjacency matrix $A \in \mathbb{R}^{n\times n}$, number k of clusters to construct.
2) Compute the unnormalized Laplacian L.
3) Compute the first k eigenvectors $v_1,...,v_k$ of L.
4) Let $V \in \mathbb{R}^{n\times k}$ be the matrix containing vectors $v_1,...,v_k$ as columns.
5) For $i = \{1,...,n\}$, let y_i correspond to the i-th row of V.
6) Cluster the points $(y_i)_i$ in $\mathbb{R}^k$ with the k-means algorithm into clusters $\tilde{C}_1,...,\tilde{C}_k$.
7) Outputs are clusters $C_1,...,C_k$ with:

$$C_i := \left\{ j : y_j \in \tilde{C}_i \right\}. \tag{3.12}$$

3.4 k-Means Algorithm

Let us cluster data D with n points and let $C_1,...,C_k$ be k disjoint clusters with $C_i \cap C_j = \varnothing$ for all $i \neq j$, where $\cup_i C_i = \{1,...,n\}$.

Definition 3.4.1. *We define the **error function** for clusters C_i with mean (or mode) $\mu(C_i)$ as:*

$$E := \sum_{i=1}^{k} \sum_{x \in C_i} d(x, \mu(C_i)). \tag{3.13}$$

Proposition 3.4.2 (k-Means Algorithm)**.** *We have the following algorithm:*
1) Input: Data set D, number of clusters k and dimensions d.
2) Initialization: $\{C_1,...,C_k\}$ is the initial partition of D.
3) Repeat:
- *Calculate $\sigma_{ij} = d(x_i, \mu(C_j))$ the distance between case i and cluster j with mode $\mu(C_j)$.*
- *Determine $z_i = \arg\min_{1\leq j\leq k} \sigma_{ij}$.*
- *Assign case i to cluster z_i.*
- *Recompute $\{\mu(C_i)\}$.*

4) Terminates when assignments are not changing.

Now recall that we clustered on the unnormalized Laplacian matrix $L = \mathrm{diag}(d_1,...,d_n) - A$ where we set $D = \mathrm{diag}(d_1,...,d_n)$. The disadvantage of this is that the degree magnitude has a large impact.

Definition 3.4.3. *We define the **normalized Laplacian** matrix $\mathcal{L}$ as:*

$$\mathcal{L} := D^{-1/2} L D^{-1/2}. \tag{3.14}$$

Normalizing the adjacency matrix makes its largest eigenvalue 1. In fact, if we let $\alpha_1 \geq ... \geq \alpha_n$ be the

eigenvalues of $\mathcal{A} = D^{-1/2}AD^{-1/2}$ and $\lambda_1 \geq ... \geq \lambda_n$ be the eigenvalues of $\mathcal{L}$, then:

$$1 = \alpha_1 \geq ... \geq -1, \quad 0 = \lambda_1 \leq ... \leq \lambda_n \leq 2. \tag{3.15}$$

Theorem 3.4.4. *For a graph G on n vertices, we have:*
1) The eigenvalues of $\mathcal{L}$ namely $\{\lambda_i\}$ satisfy:

$$\sum_{i=1}^{n} \lambda_i \leq n, \tag{3.16}$$

where exact equality holds if and only if there are no isolates, e.g. $d_i > 0$ for all i.
2) For graphs on at least two nodes $\lambda_2 \leq \frac{n}{n-1}$ with equality if and only if G is the complete graph. For a graph without isolates $\lambda_n \geq \frac{n}{n-1}$.
3) For a graph which is not a complete graph, $\lambda_2 \leq 1$.
4) If G is connected, then $\lambda_2 > 0$. If $\lambda_{i+1} = 0$ and $\lambda_{i+2} \neq 0$ then G has exactly $(i+1)$ components.
5) For all $i \neq n$ we have $\lambda_i \neq 2$ with $\lambda_n = 2$ if and only if a connected component of G is bipartite and non-trivial.

Let C_i be entities in $\{1, ..., n\}$ that belong to cluster i. Now let $z_i = s$ if $i \in C_s$ where C_s is defined from spectral clustering.
There are various modifications of k-means, which leads to the estimated labels $\widehat{z}_i$. For a choice of z such as $z = \widehat{z}$:

$$\widehat{\theta}_{ab}(z) = \frac{\sum_{i<j} A_{ij} \mathbb{1}_{\{z_i=a\}} \mathbb{1}_{\{z_j=b\}}}{\sum_{i<j} \mathbb{1}_{\{z_i=a\}} \mathbb{1}_{\{z_j=b\}}}, \quad 1 \leq a < b \leq k. \tag{3.17}$$

Denoting by $\mathcal{Z}_k$ the set of all labellings, one can also define the combinatorial least squares problem:

$$\widehat{z} = \min_{z \in \mathcal{Z}_k} \sum_{i<j} \Big(A_{ij} - \widehat{\theta}_{ab}(z) \Big)^2 \mathbb{1}_{\{z_i=a\}} \mathbb{1}_{\{z_j=b\}}. \tag{3.18}$$

Least squares is normally computationally efficient, however the addition of z makes this intractable. We finally present some comments on the degree matrix D. Although we have discussed the advantage of clustering the normalized Laplacian, the multiplication of $D^{-1/2}$ can be numerically problematic if we have some degrees that are relatively very small. For that reason, it has been suggested that the degree matrix should be normalized to:

$$D_\tau = D + \tau \mathbb{1}_n, \tag{3.19}$$

where $\tau > 0$ has to be chosen so that we have the correct balance between τ and the degree matrix. We often choose τ to be the average degree $\overline{d} = (1/n)\sum_i d_i$.
This process resembles Ridge Regression in Classical Statistics. The degree matrix D can also be regularized by an informative matrix, where again, one hopes that this moves the eigenvalues up to higher values.

3.5 Graph Limit Functions

We recall Aldous-Hoover Theorem which tells us that an infinite symmetric array $A = (A_{ij})_{i<j}$ is exchangeable if and only if there is a function $f_\alpha(x, y)$ for uniform random variable α, independent

random variables ξ_i and uniform random matrix χ_{ij} such that:

$$A_{ij}|\{\alpha,\xi\} \sim \mathbb{1}_{\{\chi_{ij}\leq f_\alpha(\xi_i,\xi_j)\}}. \tag{3.20}$$

We can note that:

$$\mathbb{P}(\chi_{ij} \leq f_\alpha(\xi_i,\xi_j)) = f_\alpha(\xi_i,\xi_j), \tag{3.21}$$

implies that:

$$\mathbb{E}[A_{ij}|\{\alpha,\xi\}] = \mathbb{P}(\chi_{ij} \leq f_\alpha(\xi_i,\xi_j)) = f_\alpha(\xi_i,\xi_j). \tag{3.22}$$

If we only observe one A then we cannot estimate the distribution of α, however we can estimate $f_\alpha(x,y)$. Now, if we imagine that A is generated by a smooth (Lipschitz) graphon f, we want to know how we could estimate f.
We could, for instance, pick a set of block widths $\{h_a\}$ so that $\sum_a h_a = n$ and fit the SBM parameters θ_{ab}. We then obtain a function estimate using $H_a := \sum_{a'=1}^{a} h_{a'}$ with initial condition $H_0 = 0$:

$$\widehat{f}(x,y) = \sum_{a,b} \widehat{\theta}_{ab} \mathbb{1}_{\{H_{a-1}<nx\leq H_a\}} \mathbb{1}_{\{H_{b-1}<ny\leq H_b\}}. \tag{3.23}$$

This produces a block constant approximation to $f(x,y)$. But how good is it? To answer this question, we need to define measure preserving transformations.

Definition 3.5.1. *We define a measure preserving map $\phi : [0,1] \longrightarrow [0,1]$ and a new function $f^\phi(x,y)$ as a **measure preserving transformation** given by:*

$$f^\phi(x,y) = f(\phi(x),\phi(y)). \tag{3.24}$$

3.6 Properties of Block Models and Generalized Wheels

For the SBM, we recall that:

$$\mathbb{P}(A_{ij} = 1|z_i = r, z_j = s) = \theta_{rs}. \tag{3.25}$$

Therefore, we can find the marginal probabilities of the model.

Proposition 3.6.1. *For a SBM with parameter θ_{rs} and h_r the number of elements in group r, the marginal probability is given by:*

$$\mathbb{P}(A_{ij} = 1) = \sum_{r=1}^{k}\sum_{s=1}^{k} \theta_{rs}\frac{h_r h_s}{n^2}. \tag{3.26}$$

Proof. We use the Total Law of Probabilities to find:

$$\begin{aligned}\mathbb{P}(A_{ij} = 1) &= \sum_{r=1}^{k}\sum_{s=1}^{k} \mathbb{P}(A_{ij} = 1, z_i = r, z_j = s) \\ &= \sum_{r=1}^{k}\sum_{s=1}^{k} \mathbb{P}(A_{ij} = 1|z_i = r, z_j = s)\mathbb{P}(z_i = r, z_j = s).\end{aligned}$$

Since we consider h_r to be the number of elements in group r, we have:

$$\mathbb{P}(A_{ij}=1)=\sum_{r=1}^{k}\sum_{s=1}^{k}\theta_{rs}\mathbb{P}(z_i=r,z_j=s)=\sum_{r=1}^{k}\sum_{s=1}^{k}\theta_{rs}\frac{h_r}{n}\frac{h_s}{n}$$
$$=\sum_{r=1}^{k}\sum_{s=1}^{k}\theta_{rs}\frac{h_r h_s}{n^2}.$$

□

Remark 3.6.2. *Remark that if $\theta_{rs}\leq\theta_{max}$ and $\theta_{min}\leq\theta_{rs}$ then $\theta_{min}\leq\mathbb{P}(A_{ij}=1)\leq\theta_{max}$.*

If we are given that $\{A_{ij}=1,A_{jk}=1\}$, we say that a triangle is completed if $\{A_{ik}=1\}$.

Proposition 3.6.3. *For a SBM with parameter θ_{rs}, the probability of triangle completion is given by:*

$$\mathbb{P}(\textit{Triangle is Completed})=\sum_{r,s,q=1}^{k}\theta_{rq}\mathbb{P}(z_i=r,z_j=s,z_k=q|A_{ij}=1,A_{jk}=1). \tag{3.27}$$

Proof. It is a straightforward calculation using the Law of Total Probabilities:

$$\begin{aligned}\mathbb{P}(\text{Triangle is Completed})&=\mathbb{P}(A_{ik}=1|A_{ij}=1,A_{jk}=1)\\&=\sum_{r,s,q=1}^{k}\mathbb{P}(A_{ik}=1,z_i=r,z_j=s,z_k=q|A_{ij}=1,A_{jk}=1)\\&=\sum_{r,s,q=1}^{k}\mathbb{P}(A_{ik}=1|z_i=r,z_j=s,z_k=q,A_{ij}=1,A_{jk}=1)\\&\qquad\times\mathbb{P}(z_i=r,z_j=s,z_k=q|A_{ij}=1,A_{jk}=1)\\&=\sum_{r,s,q=1}^{k}\theta_{rq}\mathbb{P}(z_i=r,z_j=s,z_k=q|A_{ij}=1,A_{jk}=1).\end{aligned}$$

□

These are weighted averages of θ_{ab}. If two nodes prefer links to nodes of the same type $\theta_{aa}>\theta_{ab}$, we will see more triangles than we would otherwise expect looking only at the overall density.

Special shapes can be used to retrieve the parameters of the SBM. An important example is the notion of a wheel, where we first present the definition of a wheel and then consider its generalization.

Definition 3.6.4. *A $(\mathbf{k},\mathbf{l})$-**wheel** is a graph with $kl+1$ vertices and kl edges isomorphic to the graph with edges $\{(1,2),...,(k,k+1);(1,k+2),...,(2k,2k+1);...,(1,(l-1)k+2),...,(lk,lk+1)\}$.*

Definition 3.6.5. *A **generalized** $(\mathbf{k},\mathbf{l})$-**wheel**, where $k=(k_1,...,k_t)$, $l=(l_1,...,l_t)$ are vectors and the k_j's are distinct integers, is the union $R_1\cap...\cap R_t$, where for $j=\{1,...,t\}$, R_j is a (k_j,l_j)-wheel and the wheels $R_1,...,R_t$ share a common hub but all their spokes are disjoint.*

We can now present a theorem relating the notion of a wheel with the identification of the parameters of a SBM. Let $\{W_{kl} : 1 \leq l \leq 2K-1, 2 \leq k \leq K\}$ be the specified set of (k,l)-wheels. Furthermore, let $\tau_{kl} = \rho^{-kl}\mathbb{P}(W_{kl})$ and consider A_{ij} to be generated by the sparse random graphon.

Theorem 3.6.6. *Suppose $\theta = (\pi, S)$ defines a block model with known K, and the vectors $\pi, F\pi, ..., F(K-1)\pi$ are linearly independent. Suppose $\epsilon \leq \lambda_n = \mathbb{E}(d_i) = o(\sqrt{n})$.*
Then, $\{\tau_{kl} : l = \{1, ..., 2K-1\}, k = \{2, ..., K\}\}$ identify the $\frac{K(K+3)}{2} - 2$ parameters of the block model other than ρ (i.e. the map is one-to-one).

Chapter 4

Metrics and Norms for Networks

We have specified a model for $\mathbb{E}[A_{ij}] = P_{ij}$ and have sought to estimate that model. The problem is that we have a different model as n changes. Hence we now want to find a way to compare graphs of different sizes. To achieve this, we need to define a notion of distance on graphs using metrics, where the idea is to construct an infinite dimensional model that adjusts to the size of the observations.

Definition 4.0.1. *A **metric** $d : X \times X \to [0, \infty)$ satisfies:*
1) $d(x, y) \geq 0$,
2) $d(x, y) = 0 \iff x = y$,
3) $d(x, y) = d(y, x)$ *(symmetry),*
4) $d(x, y) \leq d(x, z) + d(z, y)$ *(subadditivity).*

Definition 4.0.2. *A **kernel** on Ω is a (measurable) symmetric function $f : \Omega^2 \longrightarrow [0, 1]$.*

Definition 4.0.3. *A **graphon** is a symmetric, measurable function $f : [0, 1]^2 \longrightarrow [0, 1]$.*

Definition 4.0.4. *Let G be a simple, undirected graph. We define the **empirical graphon** $\widehat{f}^{(e)}(x, y)$ as:*

$$\widehat{f}^{(e)}(x, y) := A_{\lceil nx \rceil, \lceil ny \rceil}. \tag{4.1}$$

We remark that the empirical graphon is a step function. We recall that a function f on Ω is simple or a step function if there is a finite partition $\Omega = \cup_i B_i$ such that f is constant on each $B_i \times B_j$. Step functions are also said to be of finite type.
We now want to get to metrics/distances on f. We first start with metrics on G, where the notation W will be sometimes used for kernel functions.

4.1 Edit and Hamming Distances

We start by defining distances on two graphs, G and G', with the same node set $\{1, ..., n\}$. We can then define the following distance:

Definition 4.1.1. *With a choice of matrix norm $\|\cdot\|_M$, we define the **edit distance** to be:*

$$d_1(G,G') := \frac{\|A_G - A_{G'}\|_M}{n^2}. \tag{4.2}$$

We note that if $\|\cdot\|_M = \|\cdot\|_1 := \max_{1\leq j\leq n}(\sum_{i=1}^{n}|a_{ij}|)$ (i.e. the maximum absolute column sum), then we count the number of flipped edges times two and then divide by the total number of possibilities.

Definition 4.1.2. *With a choice of matrix norm $\|\cdot\|_M$, we define the **Hamming distance** to be:*

$$\|A_G - A_{G'}\|_M. \tag{4.3}$$

Example 4.1.3. *Suppose we observe two graphs G_1 and G_2 with the following adjacency matrices:*

$$A_1 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{4.4}$$

Hence, the edit distance is: $d_1(G_1,G_2) = \frac{\|A_1 - A_2\|_M}{n^2} = \frac{2}{16} = \frac{1}{8}$.

We further remark that sometimes the matrix norm is divided by $n(n-1)$ as in networks without self-loops the diagonal is set to zero. The Hamming distance treats as uniform all changes in the graph structure, whether they are addition in edges or deletions.
If edges are flipped with probability p then disappearances of edges follows a binomial distribution with mean $p|e(G)|$, and so do appearances.
The Hamming distance tends to place nested graphs at a smaller distance to each other than other metrics. This distance is thus a measure of the amount of change between two graphs.

4.2 Cut Norm and Cut Distance

Given a graph G and two subsets $S,T \subseteq [n]$ of its vertices, we consider $e_G(S,T)$ to be the number of edges connecting S and T.

Definition 4.2.1. *We define the **Cut norm** of G to be:*

$$\|G\|_\square := \max_{S,T\subseteq[n]} \frac{|e_G(S,T)|}{n^2}. \tag{4.5}$$

Definition 4.2.2 (Cut Distance for Finite Graphs)**.** *We define the **Cut distance for finite graphs** between two n-node graphs G_1,G_2 to be:*

$$d_\square(G_1,G_2) := \max_{S,T\subseteq[n]} \frac{|e_{G_1}(S,T) - e_{G_2}(S,T)|}{n^2}. \tag{4.6}$$

Note that if G_1 and G_2 are unlabelled, then we need to minimize $d_\square(G_1, G_2)$ over all vertex relabelling of G_1 (or G_2).

Definition 4.2.3 (Cut Distance for Graphons)**.** *Consider the group $\mathcal{S}$ of measure-preserving bijections $\phi : [0,1] \to [0,1]$. We define the **Cut distance for graphons** between two graphons W_1,W_2 to be:*

$$\delta_\square(W_1, W_2) := \inf_{\phi \in \mathcal{S}} \sup_{S,T \subseteq [0,1]} \left| \int_{S \times T} [W_1(x,y) - W_2(\phi(x), \phi(y))] \, dxdy \right|. \tag{4.7}$$

Definition 4.2.4. *We call kernels W_1 and W_2 **weakly isomorphic** if $\delta_\square(W_1, W_2) = 0$.*

4.3 Jaccard Distance

The Hamming distance has an issue with density of edges. It then becomes natural to adapt to this density, and one option is the Jaccard distance.

Definition 4.3.1. *For two graphs G_1 and G_2, the **Jaccard distance** is given by:*

$$d_J(G_1, G_2) := \frac{\|A_1 - A_2\|_1}{\|A_1 + A_2\|_*}, \tag{4.8}$$

where $\|\cdot\|_*$ is the nuclear norm $\|A\|_* = Trace\left(\sqrt{A^H A}\right) = \sum_i \sqrt{\lambda_i(A^H A)}$, with $\lambda_i(A^H A)$ being the eigenvalues of $A^H A$, and H is the Hermitian transpose.

If we now want to relate the Hamming and Jaccard distances, we can use the Steinhaus Transform. Whilst the Hamming and Jaccard distances may measure changes, such measures appear short-sighted. We use spectral matrices to cluster, where we consider the l_p distances on the eigenvalues.
If we let $\{\lambda_p\}$ be the eigenvalues of A_1 and $\{\tilde{\lambda_p}\}$ be the eigenvalues of A_2 then:

$$d(G_1, G_2)^p = \sum_{i=0}^{n-1} \left| \lambda_i - \tilde{\lambda}_i \right|^p. \tag{4.9}$$

4.4 Spanning Tree Dissimilarity

The matrix-tree dissimilarity yields a way to compute the number of spanning trees for a connected graph. We first order the eigenvalues $\lambda_0 < ... \leq \lambda_{n-1}$ and define:

$$\mathcal{T}_G = \frac{1}{n} \prod_{i=1}^{n-1} \lambda_i. \tag{4.10}$$

Definition 4.4.1. *For a graph $G = (V, E)$, a **spanning tree** is a subgraph $\widetilde{G} = (\widetilde{V}, \widetilde{E})$ that is a tree and contains all vertices of G.*

Spanning trees depict the interconnectedness of a graph, and we can quantify the addition or deletion of an edge to making a current flow on the network.

Definition 4.4.2. *For two graphs G_1 and G_2, the **dissimilarity** between G_1 and G_2 is defined as:*

$$d_{ST}(G_1, G_2) := |\log(\mathcal{T}_{G_1}) - \log(\mathcal{T}_{G_2})|. \tag{4.11}$$

Chapter 5

Network Models Continued

In this chapter, we present several other network models along with the notions of Hypergraphs and directed networks.

5.1 Exponential Random Graph Models (ERGM)

Exponential random graph models are specifying a family of probability distributions on graphs. Depending on the application, we may model simple, loopy, multiple-edged, weighted or directed graphs. Let $\mathcal{G}_n$ be the set of all graphs on n vertices.

Definition 5.1.1. *The **Exponential Random Graph Model** is given by:*

$$\mathbb{P}(\mathcal{G} = G) = \exp\left(\sum_{i=1}^{k} \theta_i T_i(G) - c(\theta)\right), \tag{5.1}$$

where for $1 \leq i \leq k$, θ_i are real valued parameters, T_i are real valued statistics defined on $\mathcal{G}_n$ and $c(\theta)$ is a normalizing constant.

ERGMs can for example be used to model relationships in Sociology. Sometimes, T_i can be chosen to be d, the total number of edges, the number of triangles/other subgraph counts or the number of connected components in the graph.

Example 5.1.2 (Special case of the Gibbs Distribution)**.** *Consider the following statistic:*

$$\widehat{t}(F, G) = \frac{X_F(G)}{n_{\left(\frac{|V(F)|}{\mathrm{aut}(F)}\right)}}. \tag{5.2}$$

Consider the ERGM given by:

$$\mathbb{P}(\mathcal{G} = G) = \exp\left(n^2(\beta_1 \widehat{t}(K_2, G) + \beta_2 \widehat{t}(K_3, G)) - c(\beta)\right), \tag{5.3}$$

where we have the normalizing constant:

$$\exp(c(\beta)) = \sum_{G \in \mathcal{G}_n} \exp\left(n^2(\beta_1 \widehat{t}(K_2, G) + \beta_2 \widehat{t}(K_3, G))\right). \tag{5.4}$$

As this is a special case of a Gibbs distribution, β_1 is known as the *particle parameter* and β_2 is known as the *energy parameter*. The exponential model is said to be *attractive* if $\beta_2 > 0$ and *repulsive* if $\beta_2 < 0$.

Remark 5.1.3. *By raising the triangle density to an exponent $\gamma > 0$, Lubetzky and Zhao proposed a generalization:*

$$\mathbb{P}(\mathcal{G} = G) = \exp\left(n^2(\beta_1 \widehat{t}(K_2, G) + \beta_2 \widehat{t}^{\gamma}(K_3, G)) - c'(\beta)\right), \tag{5.5}$$

which enabled the model to exhibit a nontrivial structure even when $\beta_2 > 0$. When $\gamma \geq \frac{2}{3}$ then this generalized model still features the Erdos-Rényi behavior, however for $\gamma < \frac{2}{3}$ there exists regions of values of (β_1, β_2) for which a typical graph drawn from the model has symmetry breaking.

5.2 Latent Space Models

In this section we present the general notion of Latent Space Models based on the representation given by Hoff (2002). One first remark is to note that Latent Space Models are not exchangeable and hence do not have an Aldous-Hoover representation. In what follows, x_{ij} are known features.

Definition 5.2.1. *Assuming that conditionally on unobserved latent variables z_i and x_{ij}, the **Latent Space Models** are given by:*

$$\mathbb{P}(A|\boldsymbol{z}, \boldsymbol{X}, \boldsymbol{\theta}) = \prod_{i<j} \mathbb{P}(a_{ij}|z_i, z_j, x_{ij}, \theta). \tag{5.6}$$

A convenient way to parametrize $\mathbb{P}(a_{ij}|z_i, z_j, x_{ij}, \theta)$ is to use logistic regression. For a simple Bernoulli, this corresponds to parametrizing y_i in terms of its success probability p_i in terms of covariate x_i to form predictor $\eta_i = \eta(x_i)$:

$$\mathbb{P}(Y_i = 1) = \mathbb{E}(Y_i) = \mu_i = p_i \quad \text{and,} \quad g(\mu_i) = \eta_i. \tag{5.7}$$

Here, $g(\mu) = \text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right)$, and so can take any value in $\mathbb{R}$.
We consider $\eta_{ij} = \gamma + \beta^T x_{ij} - |z_i - z_j|$ and for two different nodes j, k equidistant in the latent space, their log-odds ratio is determined by the covariate and so the difference is $\beta^T(x_{ij} - x_{ik})$. In his representation, Hoff writes $d_{ij} = |z_i - z_j|$ and considers the network to be represented by d_{ij} if:

$$d_{ij} > 1 \iff a_{ij} = 0, \quad \text{and,} \quad d_{ij} < 1 \iff a_{ij} = 1. \tag{5.8}$$

This is some sense equivalent to the Geometric random graph. The network is representable if such distances can be formed, e.g. the z_i in some underlying space with some dimensionality can be

constructed. Furthermore remark that by removing the covariate, the model becomes equivalent to the Chung-Lu modelling framework.
Also due to Hoff, the above model can be generalized by replacing the logit by a probit model:

$$\mathbb{P}(A_{ij}=1)=\mathbb{E}(A_{ij})=\mu_{ij}=p_{ij}, \quad \text{with,} \quad g(\mu_{ij})=\eta_{ij}, \tag{5.9}$$

where $g(\mu)=\Phi^{-1}(\mu)$ is the inverse of the CDF of a Normal distribution and with $\eta_{ij}=\gamma+\beta^T x_{ij}+\alpha(z_i,z_j)$. Let's now look at some choices for $\alpha(z_i,z_j)$:

1) Latent class/stochastic block model: $\alpha(z_i,z_j)=\theta_{z_i,z_j}$ for $z_i\in\{1,...,k\}$ and $\Theta=(\theta_{a,b})$. This encapsulates stochastic equivalence which is a type of pattern often seen in network data in which the nodes can be divided into groups such that members of the same group have similar patterns of relationships.

2) Latent distance model: $\alpha(z_i,z_j)=-|z_i-z_j|$ for $z_i\in\mathbb{R}^k$. This model encapsulates homophily, so relationships between nodes with similar characteristics are stronger than those having different characteristics. Homophily explains transitivity ("a friend of a friend is a friend"), balance ("the enemy of my friend is an enemy") and cohesive subgroups of nodes.

3) Latent eigenmodel: $\alpha(z_i,z_j)=z_i^T\Lambda z_j$ for $z_i\in\mathbb{R}^k$ and Λ a $k\times k$ matrix. An interpretation of the latent eigenmodel is that each node i has a vector of unobserved characteristics as given by z_i. Depending on the eigenvalues of Λ, similar values of z_j will lead to contributing positively or negatively to the relationship between i and j.
$\rightarrow$ The eigenmodel with k-length vectors generalizes the latent class model.
$\rightarrow$ The eigenmodel with $(k+1)$-length vectors almost generalizes the latent distance model with k dimensional latents.

5.2.1 The Random Dot Product Graph

Let F be a d-dimensional inner product distribution with i.i.d. $X_1,X_2,...,X_n\sim F$, collected in the rows of the matrix:

$$\boldsymbol{X}=(X_1,X_2,...,X_n)^T\in\mathbb{R}^{n\times d}. \tag{5.10}$$

In this model, $X_j\in\mathbb{R}^d$ and d is presumed to be known. Furthermore, for $i\neq j$ fix with F the $\mathbb{E}(X_i^TX_j)$, and with this choice select ρ_n to be the marginal probability of having an edge.

Definition 5.2.2. *Conditionally on $\boldsymbol{X}$, the **Random Dot Product Graph** is formally given by:*

$$\mathbb{P}(\boldsymbol{A}|\boldsymbol{X})=\prod_{i<j}\left(\rho_n X_i^TX_j\right)^{a_{ij}}\left(1-\rho_n X_i^TX_j\right)^{1-a_{ij}}, \tag{5.11}$$

if we assume $0\leq\rho_n X_i^TX_j\leq\mathbb{1}\cdot\rho_n$ is not identifiable, but it is there to scale the whole set of probabilities in one go.

Now, given a graph distributed by an RDPG, we may want to estimate X_i. Note that if $\boldsymbol{X}\in\mathbb{R}^{n\times d}$ and $\boldsymbol{W}\in\mathbb{R}^{d\times d}$ is a unitary matrix then:

$$(\boldsymbol{XW})(\boldsymbol{XW})^T=\boldsymbol{XWW}^T\boldsymbol{X}^T=\boldsymbol{XX}^T. \tag{5.12}$$

As this is true, latent positions $\boldsymbol{X}$ and $\boldsymbol{XW}$ give rise to the same distribution over graphs in equation (3.21). Indeed, for an exchangeable array, by Aldous-Hoover,

$$\mathbb{P}(A_{ij} = 1|\xi) = f(\xi_i, \xi_j). \tag{5.13}$$

Imagine we define a new random variable $u_i = g^{-1}(\xi_i)$, for g a bijection, then u has a different distribution to ξ but,

$$\mathbb{P}(A_{ij} = 1|\xi) = f(g(u_i), g(u_j)) \tag{5.14}$$

for $\widetilde{f} = f(g(\cdot), g(\cdot))$, and we get a different distribution.
We note that the positive semi-definite stochastic block model can be reformulated as a RDPG. We say an RDPG with latent positions $\boldsymbol{X}$ is a SBM with k blocks if the number of distinct rows in $\boldsymbol{X}$ is k, denoted $\boldsymbol{X}(1), ..., \boldsymbol{X}(k)$. In this case, we define the block membership function $\tau : [n] \mapsto [k]$ to be a function such that $\tau(i) = \tau(j)$ if and only if $X_i = X_j$.

5.2.2 Stochastic Block Model

We consider the case of a SBM in which the block memberships of each vertex is randomly assigned. More precisely, let $\pi \in (0,1)^k$ with $\sum_k \pi_k = 1$ and suppose that $\tau(1), ..., \tau(n)$ are now i.i.d. random multinomial variables π, that is, $\mathbb{P}(\tau(i) = k) = \pi_k$ for all k.
We begin with describing the notations for the spectral decomposition of the rank d positive semi-definite matrix $\boldsymbol{P} = \boldsymbol{X}\boldsymbol{X}^T$. As $\boldsymbol{P}$ is symmetric and positive semi-definite, it has a spectral decomposition of $\boldsymbol{P} = \boldsymbol{U}_P \boldsymbol{S}_P \boldsymbol{U}_P^T$, with $\boldsymbol{U}_P$ having orthogonal columns, and $\boldsymbol{S}_P$ a diagonal matrix with non-increasing entries.
One can take the adjacency spectral embedding (ASE) of $\boldsymbol{A}$ into $\mathbb{R}^d$ by first calculating $|\boldsymbol{A}| = (\boldsymbol{A}^T\boldsymbol{A})^{\frac{1}{2}}$, and then determine its spectral embedding of $\boldsymbol{U}_A$ and $\boldsymbol{S}_A$. Our goal is to estimate the latent position matrix $\boldsymbol{X}$. Now, if the matrix $\boldsymbol{P}$ were actually observable, then the spectral embedding of $\boldsymbol{P}$, given by $\boldsymbol{U}_P \boldsymbol{S}_P^{\frac{1}{2}}$, is simply some orthogonal transformation $\boldsymbol{P}$ of $\boldsymbol{X}$. As long as we only observe $\boldsymbol{A}$ and not $\boldsymbol{P}$, we have to use that $\boldsymbol{A}$ can be viewed as a small perturbation of $\boldsymbol{P} = \boldsymbol{X}\boldsymbol{X}^T$. More precisely:

$$\left\| \boldsymbol{A} - \boldsymbol{X}\boldsymbol{X}^T \right\| = o\left(\left\| \boldsymbol{X}\boldsymbol{X}^T \right\| \right). \tag{5.15}$$

Assuming that d is known, by the Davis-Kahan Theorem, the subspace spanned by the top d eigenvectors of $\boldsymbol{X}\boldsymbol{X}^T$ is well-approximated by the subspace spanned by the top d eigenvectors of $\boldsymbol{A}$. Non-parametric summaries calculable from $\boldsymbol{X}\boldsymbol{X}^T$ can therefore be calculated from $\boldsymbol{A}$.

5.3 β-Model

We have already looked at the Chung-Lu model which sets $\mathbb{E}(A_{ij}) = p_{ij} = \min(\pi_i \pi_j, 1)$. However, even if this has a closed form estimate from d, it does not naturally stay in the permitted range. The natural way out of this problem is to find a simpler way to parametrize p_{ij}. Instead of defining n parameters $\pi_i \in [0,1]$, we may consider the following model:

Definition 5.3.1 (**β-Model**). *Consider $\beta_i \in \mathbb{R}$ and set:*

$$p_{ij} = \frac{e^{\beta_i+\beta_j}}{1+e^{\beta_i+\beta_j}}. \tag{5.16}$$

In terms of the log-odds ratio, for $1 \leq i < j \leq n$, we then have:

$$\log\left(\frac{p_{ij}}{1-p_{ij}}\right) = \beta_i + \beta_j. \tag{5.17}$$

The magnitude and sign of β_i quantifies the natural tendency of node i to have ties. The degree of node i is expected to be large (small) if β_i is positive (negative).
One might be interested in estimating the parameters of this model. In fact, for $i = \{1, ..., n\}$, we have that the maximum likelihood estimate (MLE) $\widehat{\beta}_i$ satisfies:

$$d_i = \sum_{j \neq i} \frac{e^{\widehat{\beta}_i+\widehat{\beta}_j}}{1+e^{\widehat{\beta}_i+\widehat{\beta}_j}}. \tag{5.18}$$

Of course, one could question the existence, uniqueness and accuracy of the MLE. However, after carefull considerations, one can observe that for $\{d_1, ..., d_n\}$ the degree sequence of G generated from the β-model, and letting $L = \max|\beta_i|$, then there is a constant $C(L)$ such that with probability at least $1 - C(L)n^{-2}$ there exists a unique solution of the maximum likelihood equations that satisfies:

$$\max_{1 \leq i \leq n} |\widehat{\beta}_i - \beta_i| \leq C(L)\sqrt{\frac{\log(n)}{n}}. \tag{5.19}$$

5.4 Variants of Block Models

We have observed the general utility of the Stochastic Block Model, however many human interactions are more complex than a SBM would give. In the beginning of these notes, we have introduced the mixed membership stochastic block model. We generated latent variables $\boldsymbol{\xi}_i$ for each node i from the Dirichlet distribution of dimension k with parameters $\boldsymbol{\alpha}$, and defined the matrix $\boldsymbol{\Theta} = (\theta_{pq})$ where we have drawn:

$$A_{ij}|\{\boldsymbol{\xi}_i, \boldsymbol{\xi}_j\} \sim \text{Ber}(\boldsymbol{\xi}_i^T \boldsymbol{\Theta} \boldsymbol{\xi}_j). \tag{5.20}$$

This model allowed individuals to not behave only like a single group. Now, there are many other generalizations of the SBM, namely the degree-corrected SBM, the hierarchical SBM and the geometric SBM. We present one variant of the SBM, where we consider the expectation of the adjacency matrix to be P.

Definition 5.4.1. *An **overlapping SBM** assumes that P admits the representation given by:*

$$P = \rho_n \boldsymbol{Q} \boldsymbol{\Xi}^T \boldsymbol{\Lambda} \boldsymbol{\Xi} \boldsymbol{Q}, \tag{5.21}$$

for $\boldsymbol{\Xi}$ a $k \times n$ matrix where each row denotes a position in a latent space, $\boldsymbol{\Lambda}$ denotes a diagonal matrix and $\boldsymbol{Q}$ is an $n \times n$ matrix. Furthermore, ρ_n and all entries of the diagonal matrix $\boldsymbol{\Lambda}$ are non-negative.

In the above representation, $\boldsymbol{\xi}_i$ denotes the community membership of node i.

5.5 Network Sampling

We briefly present several ways to sample a network.

• Relational/Edge sampling: Relational sampling corresponds to sampling exactly relations or edges. This could, for example, be sampling phone calls. In many network applications, the relations are the primitive objects, and the vertices are derivative from these.
• Hyperedge sampling: Sampling academic articles from a research repository involves more than one actor in every relationship. Then every article represents a hyperedge.
• Path sampling: In the early days of network science, it was thought that one could ascertain network topology by analyzing the paths traversed when sending information from one part of the Internet to another. Traceroute is an example of such a method.
The high level realization is that we rarely see the entire network.
• Snowball sampling: In this method, we see a network by initializing the sampling mechanism at $N(s,0) = \{s\}$ by sampling the node s, which could be taken to be random. We then sample the set of students $N(s,1) = \{s' : A_{ss'} = 1\}$. And then we sample:

$$N(s,2) = \bigcup_{s' \in N(s,1)} \{s'' : A_{s's''} = 1\} \backslash \{N(s,0) \cup N(s,1)\}. \tag{5.22}$$

Iterating this process for general s,k we sample:

$$N(s,k) = \bigcup_{s' \in N(s,k-1)} \{s'' : A_{s's''} = 1\} \backslash \left\{ \bigcup_{j=0}^{k-1} N(s,j) \right\}. \tag{5.23}$$

Then, the snowball sampling is the set of neighborhoods $\{N(s,k)\}_{0 \le k \le r}$.

It used to be standard to think that if you observed one network then you had one observation. And in fact, we can bring the notion of large samples to this idea by instead considering many observational units. The sample size is the number of observational units. In theory, we could view a network in terms of the number of observational units/edges with $\binom{n}{2}$ of them. Yet, this does not reflect the reality when the network is sparse, where we consider the number $\rho\binom{n}{2}$ to be more realistic.

5.6 Barabasi-Albert Model

In this section, we introduce one more model that is partially non-statistical. The Barabasi-Albert preferential attachement Model was introduced to explain a number of empirical characteristics observed in real networks.
A network evolves in this model by adding a vertex at a time with each new vertex attaching to existing vertices according to their degree. We assume that there are t vertices at time t, we fix $m \in \mathbb{N}$, and write the degree of vertex $v_i^{(m)}$ as $d_i(t)$.
To map the network at time point t to that at time point $t+1$, we add a new vertex to the network that has m edges. We now need to decide how these connect to the existing vertices. We take $\delta > -m$

and at each time step a new vertex appears and attaches at random to m existing vertices with a probability proportional to the degree of that vertex shifted to be offset by δ.

Definition 5.6.1. *The **Barabasi-Albert Model** corresponds to the case where $m=1$, and the probability that there is an edge between nodes i and $t+1$ is (self-loops admitted):*

$$\mathbb{P}(v_i^{(1)} \text{ connects to } v_{t+1}^{(1)}) = \begin{cases} \frac{d_i(t)+\delta}{t(2+\delta)+1+\delta}, & \text{for } i = t+1 \\ \frac{1+\delta}{t(2+\delta)+1+\delta}, & \text{for } i \in \{1,...,t\} \end{cases}. \tag{5.24}$$

For $m>1$, the network is constructed from a Preferential Attachment Model with $m=1$ and $\delta' = \delta/m$. We start with some existing graph G_0 and keep going for G_1, G_2, etc..., where we assume that G_0 has k_0 edges. Normally networks are sparse and having few very large degrees.

Definition 5.6.2. *Consider $\{a_{ij}^{(n)}\}$ to be the collection of edge variables at time step n. A network is said to be **sparse** if:*

$$\lim_{n\to\infty} \frac{2}{n(n-1)} \sum_{i<j} a_{ij}^{(n)} = \lim_{n\to\infty} \widehat{\rho} = 0. \tag{5.25}$$

In other words, a network is sparse if and only if $\frac{\#\text{edges}}{\#\text{possible edges}} \to 0$ as $n\to\infty$. By the generating dynamics of the Barabasi-Albert Model, there are m new edges at each step so $k_n = mn + k_0$. Thus:

$$\widehat{p} = \frac{2(mn+k_0)}{n(n-1)} \to 0. \tag{5.26}$$

The degree distribution counts the relative proportion of vertices of any integer degree:

$$p_{A^{(n)}}(k) = \sum_{i=1}^{n} \mathbb{1}_{\{d_i^{(n)}=k\}}, \tag{5.27}$$

and $p_{A^{(n)}}(k) \sim k^{-\gamma}$ corresponds to a power law distribution.

5.7 Statistical Temporal Models

The generating mechanism of the Preferential Attachment Model is very mechanistic. What if we want to generate a set of edges (or contacts) over time? We could simply start from the graph limit model:

$$\mathbb{E}(A_{ij}|\xi) = \rho_n f(\xi_i, \xi_j). \tag{5.28}$$

We shall assume we observe multiple graphs across time and so for a given edge ij we study the edge variable $A_{ij}(t)$. The simplest model takes the form:

$$\mathbb{P}(A_{ij}(t) = 1 | A_{ij}(t-1), ...) = h(t, A_{ij}(t-1), A_{ij}(t-2), ...), \tag{5.29}$$

for some appropriately chosen function $h(\cdot)$.

We shall now look at how the edges can be modeled across times with label vector z:

$$A_{ij}(t)|\{A_{ij}(t-1),...,A_{ij}(t-k)\} \sim \text{Ber}\left(\text{logit}^{-1}\left(\sum_{\alpha=1}^{k} b_{z_i z_j \alpha} A_{ij}(t-k) + c_{z_i z_j}(t)\right)\right). \tag{5.30}$$

This model defines a correlated process across time, where the correlation is specified by b. It uses the trick of utilizing a modelling framework that naturally limits the success probabilities between 0 and 1. We present below a list of properties of such models:
→ This model introduces flexible forms of serial correlation.
→ The generating mechanism can still be estimated and the properties of the network determined.
→ Non-stationary models can be included by replacing c_{ij} by time-varying alternatives as we have presented above.
→ Parameters naturally stay within the range of allowed values.
→ The simplest version of this model takes $k = 1$, which introduces series correlation of length 1.
→ This model is different from just letting the parameters of the SBM change over time.

5.8 Link Prediction

What is the point of link prediction? Suppose we only know a portion of the adjacency matrix (or possibly we don't know the distribution of the graph), and we want to infer on the missing values. It is very difficult to predict whether an edge is actually absent or if we are just missing some informations. Link prediction will let us study such models where we are missing some informations.
In this section, let $G = (V, E)$ be an undirected network, where we assume that all vertices $v \in V$ are known. We let $V^{(2)}$ be the set of distinct unordered edges. We will use the term "edge" to signify an interaction between i and j, the term "edge variable" for the knowledge of whether an interaction does or does not exist, and finally "missing information/edge variable" if we don't know if an edge exists. In this section we will suppose the presence or absence of an edge in $V^{(2)}$ is only observed for $V^{(2)}_{\text{obs}}$, and we consider that the information is missing for $V^{(2)}_{\text{miss}} = V^{(2)} \backslash V^{(2)}_{\text{obs}}$. We furthermore let $\mathbf{A}$ be a random $n \times n$ matrix with binary entries. We denote by $\mathbf{A}^{\text{obs}}$ and $\mathbf{A}^{\text{miss}}$ the adjacency matrices corresponding to $V^{(2)}_{\text{obs}}$ and $V^{(2)}_{\text{miss}}$.

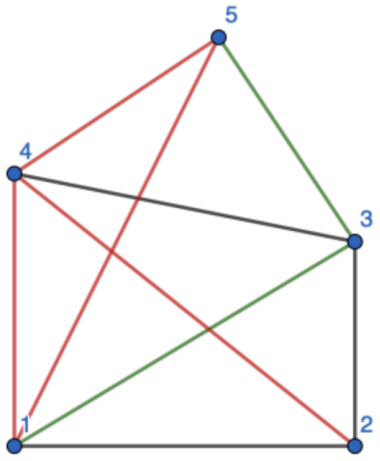


Figure 5.1: Black edges are usual edges, red edges are absent edges and green edges are missing informations.

The problem of Link Prediction is to predict $\mathbf{A}^{\text{miss}}$ given $\mathbf{A}^{\text{obs}} = \mathbf{a}^{\text{obs}}$ as well as possible vertex attributes $\mathbf{X} = \mathbf{x} \in \mathbb{R}^n$. Often, missingness is due to sampling, and we must model the sampling mechanism. Also, it is usually assumed that edge variables are missing at random, where this assumption corresponds to the condition of whether A_{ij} is observed depends only on whether other $A_{i'j'}$ are observed, not on the variable A_{ij} itself. If it were the case that the probability that A_{ij} was missing depended on the value of A_{ij} then this is called informative missingness.
Given a suitable model for $\mathbf{X}$ and $(\mathbf{A}^{\text{miss}}, \mathbf{A}^{\text{obs}})$ we might seek:

$$\mathbb{P}\left(\mathbf{A}^{\text{miss}} | \mathbf{A}^{\text{obs}} = \mathbf{a}^{\text{obs}}, \mathbf{X} = \mathbf{x}\right). \tag{5.31}$$

5.9 Scoring Methods

For each pair i and j whose edge status is unknown, i.e. $\{i,j\} \in V^{(2)}_{\text{miss}}$, a score value $s(i,j)$ is computed. Normally, a set of predicted edges are found by setting a threshold $s^* > 0$ and then keeping the edges for which $s(i,j) > s^*$. A number of scores have been proposed. Scores are generally chosen to determine certain structural characteristics of a network graph $G^{\text{obs}} = (V^{\text{obs}}, E^{\text{obs}})$ associated with $\mathbf{A}^{\text{obs}} = \mathbf{a}^{\text{obs}}$. A simple choice would be:

$$s(i,j) = -\text{dist}_{G^{\text{obs}}}(i,j), \tag{5.32}$$

where the negative sign in this equation is included so that larger values indicate that vertices have increased probability of sharing an edge.
There are also a number of scores based on the 1-neighbourhood of node i, $N(i,1)$. Set $N^{\text{obs}}(i,1)$ and we have the following definition:

Definition 5.9.1. *The **neighborhood score** is defined as:*

$$s(i,j) := \left|N^{obs}(i,1) \cap N^{obs}(j,1)\right|. \tag{5.33}$$

This assumes that in the missing data, edges are more likely if observed edges exist between the nodes. As n grows in size this would naturally grow and so a normalized version is given by the following:

Definition 5.9.2. *The **Jaccard coefficient score** is defined as:*

$$s(i,j) := \frac{\left|N^{obs}(i,1) \cap N^{obs}(j,1)\right|}{\left|N^{obs}(i,1) \cup N^{obs}(j,1)\right|}. \tag{5.34}$$

This also relies on the observed and missing data having the same characteristics. A score relating to the Jaccard Coefficient Score is the Liben-Nowell Score:

$$s(i,j) := \sum_{k \in N^{\text{obs}}(i,1) \cap N^{\text{obs}}(j,1)} \frac{1}{\log |N^{\text{obs}}(k,1)|}. \tag{5.35}$$

This score weights heavily the common neighbours of i and j that are not themselves highly connected. A problem with those scores is that they are heavily local. The implicit assumption to this choice is that vertices in the observed graph are tied implies that they should be linked also in the missing data.

Link prediction can be approached as binary classification. Taking $\mathbf{A}^{\text{obs}} = \mathbf{a}^{\text{obs}}$ as training data, our goal is to use the data $\mathbf{a}^{\text{obs}}$ and any vertex attributes $\mathbf{X}$ to construct a classifier. Then we use the classifier to predict entries in $\mathbf{A}^{\text{miss}}$. We attempt to "approximate" the rule that determines the presence of edges using Logistic Regression.

5.10 Logistic Regression

We shall explain what we choose as $\mathbf{z}$ momentarily. The model will be of the form:

$$\log\left(\frac{\mathbb{P}_{\boldsymbol{\beta}}(A_{ij}=1|\boldsymbol{Z}_{ij}=\boldsymbol{z})}{\mathbb{P}_{\boldsymbol{\beta}}(A_{ij}=0|\boldsymbol{Z}_{ij}=\boldsymbol{z})}\right)=\boldsymbol{\beta}^T\boldsymbol{z}. \tag{5.36}$$

Here, $\boldsymbol{Z}_{ij}$ is a vector of explanatory variables indexed by the pair $\{i,j\}$. Estimates of $\boldsymbol{\beta}$ can be produced using maximum likelihood. Potential edges (i,j) are classified as present or absent according to estimated classification probabilities:

$$\mathbb{P}_{\widehat{\boldsymbol{\beta}}}\left(A_{ij}^{\text{miss}}=1|\boldsymbol{Z}_{ij}=\boldsymbol{z}\right)=\frac{\exp\left(\boldsymbol{\beta}^T\boldsymbol{z}\right)}{1+\exp\left(\boldsymbol{\beta}^T\boldsymbol{z}\right)}, \tag{5.37}$$

and if this quantity exceeds a threshold of $\frac{1}{2}$, the edge is deemed present. The $\boldsymbol{Z}_{ij}$ are here vectors of functions:

$$\boldsymbol{Z}_{ij}=\left(g_1\left(\boldsymbol{A}_{(-ij)}^{\text{obs}},\boldsymbol{X}\right),...,g_k\left(\boldsymbol{A}_{(-ij)}^{\text{obs}},\boldsymbol{X}\right)\right)^T, \tag{5.38}$$

where $\boldsymbol{A}_{(-ij)}^{\text{obs}}$ is all elements of $\boldsymbol{A}^{\text{obs}}$ barring A_{ij}. We only use A_{ij} to train the classifier. The functions $g_1(\cdot),...,g_k(\cdot)$ are selected to parametrize predictive information in $\boldsymbol{A}^{\text{obs}}=\boldsymbol{a}^{\text{obs}}$ and $\boldsymbol{X}=\mathbf{x}$.
Standard logistic regression framework would have A_{ij} as independent. We do not know the effect on classifier accuracy dependence has. The underlying missingness mechanism should affect the answer. We shall think about logistic regression with latent variables.
Let $\boldsymbol{M}$ be an unknown random symmetric $n\times n$ matrix of the form:

$$\boldsymbol{M}=\boldsymbol{U}^T\boldsymbol{\Lambda}\boldsymbol{U}+\boldsymbol{E}. \tag{5.39}$$

In this representation, $\boldsymbol{U}=(\boldsymbol{u}_1,...,\boldsymbol{u}_n)$ is a random but orthogonal matrix, $\boldsymbol{\Lambda}$ is a $n\times n$ random diagonal matrix, and $\boldsymbol{E}=(\epsilon_{ij})$ is a matrix of iid noise variables. To preserve symmetry we take $\epsilon_{ij}=\epsilon_{ji}$, and the matrix $\boldsymbol{M}$ conditionally takes the form:

$$\boldsymbol{M}_{ij}|\boldsymbol{U},\boldsymbol{\Lambda},\boldsymbol{E}=\boldsymbol{u}_i^T\boldsymbol{\Lambda}\boldsymbol{u}_j+\epsilon_{ij}. \tag{5.40}$$

We augment $\boldsymbol{Z}_{ij}$ by the unobserved variable $\boldsymbol{M}_{ij}$ so we get:

$$\log\left(\frac{\mathbb{P}_{\boldsymbol{\beta}}(A_{ij}=1|\boldsymbol{Z}_{ij}=\boldsymbol{z},\boldsymbol{M}_{ij}=m)}{\mathbb{P}_{\boldsymbol{\beta}}(A_{ij}=0|\boldsymbol{Z}_{ij}=\boldsymbol{z},\boldsymbol{M}_{ij}=m)}\right)=\boldsymbol{\beta}^T\boldsymbol{z}+m. \tag{5.41}$$

With this model, if we are only given $\boldsymbol{Z}_{ij}$ then $\{A_{ij}\}$ are dependent. The dependence is a consequence of the hierarchical nature of the model.
We introduced $\boldsymbol{M}$ to capture effects that are not captured by $\boldsymbol{Z}_{ij}$. To produce statistical predictions, we need to specify distributions for $\boldsymbol{U}$, $\boldsymbol{\Lambda}$ and $\boldsymbol{E}$. For a Bayesian analysis, priors for $\boldsymbol{\beta}$ need to be

specified.
To determine if there are edges we use elements in $\boldsymbol{A}^{\text{miss}}$ we compute:

$$\mathbb{E}\left(\frac{\exp\left(\boldsymbol{\beta}^T\boldsymbol{Z}_{ij}+\boldsymbol{M}_{ij}\right)}{1+\exp\left(\boldsymbol{\beta}^T\boldsymbol{Z}_{ij}+\boldsymbol{M}_{ij}\right)}|\boldsymbol{A}^{\text{obs}}=\boldsymbol{a}^{\text{obs}},\boldsymbol{Z}_{ij}=\boldsymbol{z}\right), \tag{5.42}$$

and compare this value to a given threshold.

5.11 Biclustering

We have looked at $G=(V,E)$ as an undirected network. We shall now generalize this to having two types of nodes, V_1 and V_2 respectively. There are edges between $v_1 \in V_1$ and $v_2 \in V_2$ given by the edge set E. One might want to understand two sets of objects V_1 (e.g. consummers) and V_2 (e.g. goods). Thus, the intrinsic statistical object is (V_1,V_2,E) with $|V_1|=n$ and $|V_2|=m$. We assume that the data form has undirected edges. The edges are collected in a data matrix X. The common inference problem addressed in terms of the data matrix is usually one of clustering.
We can list several examples of biclustering applications:
• For online retail, X_{ij} can then show if user i wants product j, and our task is to segment users and products into relevant subgroups. The question would be to know who might want product j.
• In bioinformatics, X_{ij} could correspond to the log activation level of gene j in patient i. The task is then to determine groups of patients with similar genetic profiles, while at the same time finding groups of genes with similar activation levels.
• In medicine, determining groups in such data bases has helped to identify associations between active ingredients and adverse medical reactions.

Standard clustering can be applied either to rows or columns of a symmetric matrix A. Unlike A, X is not symmetric. The process of biclustering clusters both these dimensions simultaneously. Biclustering algorithms identify groups of variables that display similar activity patterns under a specific subset of the common features.
We write X_{ij} for the response of variable i under condition j. We can represent this in two ways by its set of rows $R=\left\{x_1^{(r)},...,x_n^{(r)}\right\}$ and by its set of columns $C=\left\{x_1^{(c)},...,x_m^{(c)}\right\}$. We will write $X=(R,C)$, $I \subset R$, $J \subset C$ and X_{IJ} for the submatrix that has rows I and columns J.

Definition 5.11.1. *A **cluster of rows** is a subset of rows that exhibits similar behaviour across the set of all columns.*

A row cluster $X_{IC}=(I,C)$ is a subset of rows. Here $I=\{i_1,...,i_k\}$ is a subset of rows ($I \subset R$ and $k \leq n$). A cluster of rows (I,C) can thus be defined as a $k \times m$ submatrix of the matrix X.

Definition 5.11.2. *A **cluster of columns** is a subset of columns that exhibits similar behaviour across the set of all rows.*

A column cluster $X_{RJ}=(R,J)$ is a subset of columns defined over the set of all rows R, where $J=\{j_1,...,j_s\}$ is a subset of columns ($J \subset C$ and $s \leq m$). A cluster of columns (R,J) can thus be defined as a $n \times s$ submatrix of the matrix X. In contrast a bicluster is a subset of rows that exhibits

similar behaviour across a subset of columns and vice versa. The bicluster X_{IJ} is then a subset of rows and a subset of columns where $I = \{i_1, ..., i_k\}$ is a subset of rows ($I \subset R$ and $k \leq n$), and $J = \{j_1, ..., j_s\}$ is a subset of columns ($J \subset C$ and $s \leq m$).

Definition 5.11.3. *A **bicluster** (I, J) is a $k \times s$ submatrix of the matrix X.*

Given a data matrix X we want to identify a set of biclusters $B_k = (I_k, J_k)$ so that B_k is in some sense homogeneous. Such a matrix can be viewed as a bipartite graph.
There are four types of biclusters: (i) Biclusters with constant values, (ii) Biclusters with constant values on rows or columns, (iii) Biclusters with coherent values, and (iv) Biclusters with coherent evolutions. The first three types analyse the observed values in X and the fourth tries to identify a coherent behaviour. To numerically summarize the data matrix, we define:

$$\overline{X}_{i,J} = \frac{1}{|J|} \sum_{j \in J} X_{ij}, \qquad \overline{X}_{I,j} = \frac{1}{|I|} \sum_{i \in I} X_{ij}, \qquad \overline{X}_{I,J} = \frac{1}{|I||J|} \sum_{i \in I, j \in J} X_{ij}. \tag{5.43}$$

Definition 5.11.4. *A **perfect constant bicluster** is a submatrix (I, J) where all values are equal. For all $i \in I$ and $j \in J$ we have $X_{ij} = \mu$.*

Sometimes, such ideal biclusters can be found in some data matrices. The values X_{ij} found in what can be considered as a constant bicluster are generally presented as $\eta_{ij} + \mu$, where η_{ij} is the noise associated with the real value of X_{ij}. Hartigan introduced block clustering to find these groups. This algorithm splits the original data matrix into a set of submatrices (biclusters) and uses the variance to evaluate the quality of each bicluster (I, J):

$$\text{Var}(I, J) = \sum_{i \in I, j \in J} \left(X_{ij} - \overline{X}_{I,J} \right)^2. \tag{5.44}$$

To avoid continuous partitioning , Hartigan assumes that there are K biclusters within the data matrix. The algorithm stops when the data matrix is partitioned into K biclusters and the quality of the resulting biclustering is computed as:

$$\text{Var}(I, J)_K = \sum_{k=1}^{K} \sum_{i \in I_k, j \in J_k} \left(X_{ij} - \overline{X}_{I_k, J_k} \right)^2. \tag{5.45}$$

Hartigan designed a permutation-based method to induce the optimal number of biclusters, K.

Definition 5.11.5. *A **perfect bicluster** with constant columns is a submatrix (I, J) where all the values within the bicluster can be obtained using one of the following expressions:*

$$\mathbb{E}(X_{ij}) = \mu + \alpha_i \quad \text{or,} \quad \mathbb{E}(X_{ij}) = \mu \alpha_i. \tag{5.46}$$

Here μ is the typical value within the bicluster, and α_i is the adjustment for row $i \in I$.

Definition 5.11.6. *A **perfect bicluster** with constant rows is a submatrix (I, J) where all the values within the bicluster can be obtained using one of the following expressions:*

$$\mathbb{E}(X_{ij}) = \mu + \beta_j \quad \text{or,} \quad \mathbb{E}(X_{ij}) = \mu\beta_j. \tag{5.47}$$

Here μ is the typical value within the bicluster, and β_j is the adjustment for column $j \in J$.

5.12 Extensions of Exchangeability, Graphex and Edge Exchangeability

We have noticed that permutation invariance is a natural probabilistic symmetry for networks. We can either consider finite exchangeability, or a finite sample from an infinite exchangeable array. However, exchangeability has a number of drawbacks:

- Sparsity: The simplest (vertex) exchangeable representation is not consistent with sparsity.
- Internal Heterogeneity: Neither are simple (vertex) exchangeable models easily brought together with power-law degrees.
- External Heterogeneity: It is more challenging to bring exchangeability together with covariates and other informations.

We therefore need some alternatives. Recall that the graphon Aldous-Hoover framework used a function $f(x,y)$, $[0,1]^2 \mapsto [0,1]$ to parametrise the distribution of $\{A_{ij}\}_{i>j}$. As an alternative, a network can be constructed from a point process $Y_t \subset [0,t] \times [0,t]$. We say that Y_t is exchangeable if its distribution is unchanged by applying any measure preserving transformation. Now, we observe A where A_{ij} is unity if $(\theta_i, \theta_j) \in Y$, where $\theta_1, \theta_2, ...$ are the arrival times of vertices in the point process. Note that $\{Y_t\}$ being stationary does not mean X is exchangeable. The occurence of $(t,t') \in y$ means that there is an edge between vertices labelled t and t'. It is tempting to interpret t as the time a vertex enters the system, however this is overinterpreting. But its connection to real data is easier to make with this interpretation. So now what? How do we get sparse and power-law?

Definition 5.12.1. *A **graphex** is a triplet (I, S, W) where $I \in [0,\infty)$, $S : [0,\infty) \mapsto [0,\infty)$ is a measurable map, and $W : [0,\infty)^2 \mapsto [0,1]^2$ a graphon function.*

We take realisations of a unit-rate point processes $\Xi = \{(\theta_i, \theta_j)\}$, $\Xi'_i = \{(\sigma_{ij}, \chi_{ij})\}$ and $\Xi'' = \{(\rho_j, \rho'_j, \eta_j)\}_j$. θ_i, σ_{ij} are potential vertex labels, while they can be regarded as types of the corresponding labels. The point process determines the stochastic properties of θ_i. We then realize a graph A by a suitable combination of the three random arrays. The most important edges are present with probability/conditional expectation,

$$\mathbb{P}(W(\theta_i, \theta_j) \leq \xi_{ij}), \tag{5.48}$$

where $\{\xi_{ij}\}$ are i.i.d. uniform random variables. Additional edges are present due to: (1) isolated *stars*, and (2) isolated *edges*.

But now, what if we start to observe edges rather than nodes? We assume we observe an edge list $E_n = \{E_i\}_{i \in [n]}$, and that we have a permutation $\sigma : [n] \mapsto [n]$, defining $E_n^\sigma = \{E_{\sigma(i)}\}_{i \in [n]}$. We write a random edge labeled network $Y = \{Y_i\}_{i \in \mathbb{N}}$.

Definition 5.12.2. *A randomly edge labeled network* $\mathcal{Y} = \{\mathcal{Y}_i\}_{i\in\mathbb{N}}$ *is* ***edge exchangeable*** *if* $\mathcal{Y}^\sigma = \mathcal{Y}$ *in Law, for all permutations* $\sigma : \mathbb{N} \longrightarrow \mathbb{N}$.

Edge exchangeable models have the same probability assigned to all edge labeled graphs that are equivalent up to a choice of relabeling. Any edge labeled network $\mathcal{Y} = \{\mathcal{Y}_i\}_{i\in\mathbb{N}}$ yields a compatible sequence of finite networks $\mathcal{Y} = \{\mathcal{Y}_i\}_{i\in\mathbb{N}}$ by taking $\mathcal{Y}_n = \mathcal{Y}|_n$ to be the restriction of subsampling $[n] \subset \mathbb{N}$. Any such sequence is infinitely edge exchangeable, namely $\mathcal{Y}^\sigma = \mathcal{Y}$ in Law for all permutation $\sigma : [n] \longrightarrow [n]$ and $\mathcal{Y}_n|_{[m]} = \mathcal{Y}_m$ in Law for all $n \geq m \geq 1$.
Even if all edges arrive in an exchangeable process , the vertices arrive in biased order weighted by the relative frequency of their occurence in the network interactions. Therefore, the sample of vertices can be argued that it does not represent an exchangeable draw from the population of vertices.

Chapter 6

Multidimensional Networks and Directed Networks

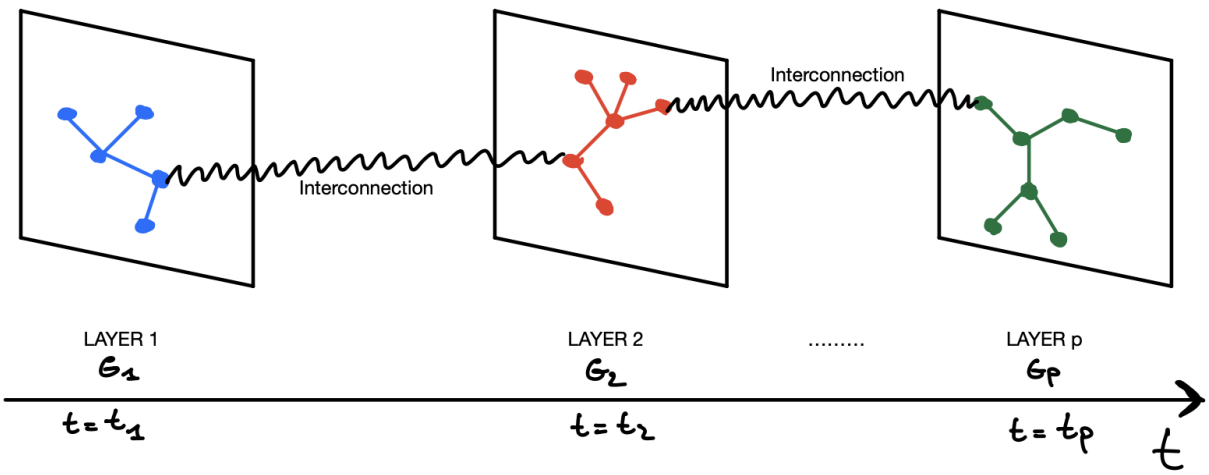
In this chapter we present two different types of multidimensional networks, corresponding to Multilayer, Multiplex and Hypergraphs. We also define Directed Networks and their properties.

6.1 Multilayer and Multiplex Networks

Definition 6.1.1. *A **multilayer network** is a pair $M = (G, C)$ where $G = \{G_p : p \in \{1, ..., P\}\}$ is a family of simple graphs $G_p = (V_p, E_p)$ called the layers of M, and C is:*

$$C = \{E_{ab} \subset V_p \times V_p : a \neq b \in \{1, ..., P\}\}, \tag{6.1}$$

is the set of interconnections between nodes of different layers G_a and G_b with $a \neq b$. The items in C are crossed layers and the elements of E_p are intralayer connections. Note that the elements of E_{ab} are interlayer connections.



This model is useful to capture phenomena in social systems, such as the Axelrod Model. A general multilayer network has potentially different vertices on different layers. We can also have edges between nodes in different layers. Of course, we can collect all edges in a supra-adjacency matrix which captures all edges present in the network, which is however very complex to construct.

A special case of the multilayer network is the multiplex network. In this model, $V_1 = ... = V_p$. Thus a multiplex network has a fixed set of nodes that are connected with different edges. Simple graphs can be defined from the multiplex network. The projection network $proj(M)$ has an adjacency matrix, for some p, given by:

$$\overline{A}_{ij} = \mathbb{1}_{\{A_{ij}^{(p)}=1\}}. \tag{6.2}$$

Multilayer networks occur from several different causes: Multiplex networks, Temporal networks, Interacting or interconnected networks, Multidimensional networks or Independent (layered) networks.

Definition 6.1.2. ***Multiplex networks*** *are several networks with the same nodes, with layers of edges.*

Definition 6.1.3. ***Temporal networks*** *are several graphs G_t that are recorded at time t.*

If we consider a family of networks $\{G_1, ..., G_L\}$ that interact and can be modelled as a multilayer network of layers. Thus we can write the adjacency matrix as $A_{ij}(t)$ say, but where the t index does not have to be time. Therefore, the observations correspond to the tensor $\{A_{ij}(t)\}$. We can define the time-varying degree vector $d(t)$, and in fact any other time-varying statistics that we may choose. Note that we can also estimate time-varying groups.

6.2 Directed Networks

Social Networks are often not symmetric, with examples including an acquaintance network of university members (Kossinets and Watts (2006)), a large scale instant messaging network containing individuals (Leskovec and Horvitz (2008)), friendship networks of a set of American high schools, or a social network of a cohort of college students in Facebook (Lewis, Gonzalez and Kaufman). Furthermore, we often have directed relationships on email and other modes of communication. Powerline networks, airline networks, citation networks, food webs and Economic networks are all directed networks.

To replicate the patterns we see in practice, we need to generalize edges to directed edges. A directed edge $\overrightarrow{ij}$ is sometimes called an arc. An arc (i, j) is considered to be directed from i to j, where j is called the head and i is called the tail of the arc. The adjacency matrix is then defined to be:

$$A_{ij} = \mathbb{1}_{\left\{\overrightarrow{ij} \text{ is present}\right\}}. \tag{6.3}$$

Notice that it follows that $A \neq A^T$. We now need two summaries for the popularity or sociability of a node. We consider the in-degree (pointing to i) to be:

$$d_i^{(\text{in})} = \sum_{j \neq i} A_{ji}, \tag{6.4}$$

and the out-degree (pointing away from i) to be:

$$d_i^{(\text{out})} = \sum_{j \neq i} A_{ij}. \tag{6.5}$$

• Furthermore, versions of Configuration Model can be defined for directed graphs with parameters $\boldsymbol{\pi}^{(\text{in})}$ and $\boldsymbol{\pi}^{(\text{out})}$. Thus the degree based model becomes:

$$\mathbb{E}(A_{ij}) = \pi_i^{(\text{in})}\pi_j^{(\text{out})}. \tag{6.6}$$

Unlike standard Configuration Model, we now have $2n$ parameters. We could for example postulate $\boldsymbol{\pi}^{(\text{in})} = \rho^{(\text{in})}\boldsymbol{\pi}$ and $\boldsymbol{\pi}^{(\text{out})} = \rho^{(\text{out})}\boldsymbol{\pi}$.

• We can also do a permutation invariant version of this degree-based model. Hence we need two 1-dimensional functions $f^{(\text{in})}(x)$ and $f^{(\text{out})}(x)$. We then define two latent independent vectors ξ and η with uniform entries and so we take:

$$\mathbb{E}(A_{ij}|\xi,\eta) = \rho_n \cdot f^{(\text{in})}(\xi_i)f^{(\text{out})}(\eta_j). \tag{6.7}$$

And we observe that we are no longer in the land of symmetric adjacency matrices.

• There is a directed version of the graphon model. Specify a graph limit function $f(x,y)$ which is no longer symmetric. Specify α as a latent variable, as well as two latent independent vectors ξ and η with uniform entries and so take,

$$\mathbb{E}(A_{ij}|\xi,\eta) = \rho_n \cdot f_\alpha(\xi_i,\eta_j), \tag{6.8}$$

and conditional on the latent variables generate the adjacency matrix as independent Bernoulli random variables.

• Some graphs display clear group structures. For each node i we define a random variable $z_i^{(\text{in})}$ that takes the value $\{1,...,k_{(in)}\}$, where this variable is indicating the in-group membership of node i. For each node i we define a random variable $z_i^{(\text{out})}$ that takes the values $\{1,...,k_{(\text{out})}\}$, where this variable is indicating the out-group membership of node i. We additionally define a connection probability matrix Θ which has entries θ_{ab} for $1 \le a \le k_{(\text{in})}$ and $1 \le b \le k_{(\text{out})}$. Then:

$$A_{ij}|\left\{z_i^{(\text{in})}, z_i^{(\text{out})}\right\} \sim \text{Ber}\left(\theta_{z_i^{(\text{in})} z_j^{(\text{out})}}\right), \quad 1 \le j \ne i \le n, \tag{6.9}$$

where each realizations are independent. Furthermore, $A_{ii} = 0$ for $1 \le i \le n$ and we complete the matrix as usual. This is known as the Directed SBM.
One can find a maximum likelihood estimation of this set of parameters, and in particular we can look at some very small examples. We can use the same type of algorithms as for regular SBM, where we will yet again use profile likelihood to estimate the parameters of the model:

$$\begin{aligned} l(\theta, z^{(\text{in})}, z^{(\text{out})}) &= \log\left(\prod_{i\ne j} \theta^{a_{ij}}_{z_i^{(\text{in})} z_j^{(\text{out})}} \left(1 - \theta_{z_i^{(\text{in})} z_j^{(\text{out})}}\right)^{1-a_{ij}}\right) \\ &= \sum_{i\ne j}\left(a_{ij}\log\left(\theta_{z_i^{(\text{in})} z_j^{(\text{out})}}\right) + (1-a_{ij})\log\left(1-\theta_{z_i^{(\text{in})} z_j^{(\text{out})}}\right)\right). \end{aligned}$$

As a last comment, having removed the assumption of symmetry, we could also start to look at relationships between different types of objects. We could therefore look at some generalizations of A_{ij} that are not square, and have ties between predators and preys for instance. We could also use the Directed SBM to estimate the graph limit without symmetry.

6.3 Hypergraphs

We now look to generalize the idea of a graph, and hypergraphs lets us consider subsets of nodes to be edges and not only pairs of nodes anymore.

Definition 6.3.1. *Let $V = \{1, ..., n\}$ be the vertex set and let $\mathcal{D} = \{D_1, ..., D_m\}$ be a family of subsets of V. A **hypergraph** is a pair $\mathcal{H} = \{V, \mathcal{D}\}$ with vertex set $V(\mathcal{H})$ and edge set $\mathcal{D}(\mathcal{H})$.*

In general we have that $|V| = n$ is the order of the hypergraph, and we usually write by $m(\mathcal{H})$ the number of edges. Furthermore we include the cases where V and $\mathcal{D}$ are empty, and if the hypergraph has no multiple or included edges then it is simple where an included edge is an edge that is a subset of other edges. Similarly to basic graphs, two vertices are adjacent if there is an edge or a hyperedge that connects them, and two adjacent vertices are neighbours. All the adjacent vertices of i are called the neighbourhood of i.

Definition 6.3.2. *If a vertex i belongs to an edge $\mathcal{D}_j \in \mathcal{D}$ then i and $\mathcal{D}_j$ are said to be **incident**.*

We write $\mathbb{D}(i)$ for $i \in V$ for all edges containing vertex i. Hence the number $|\mathbb{D}(i)|$ is the degree of vertex i. In contrast, $|\mathcal{D}_j|$ is the cardinality of edge $\mathcal{D}_j$ (which was always 2 in basic graphs).

Definition 6.3.3. *We define the **maximum degree** of a hypergraph $\mathcal{H}$ to be:*

$$\Delta(\mathcal{H}) = \max_{i \in V} |\mathbb{D}(i)|. \tag{6.10}$$

Definition 6.3.4. *A hypergraph in which all vertices have degree $k \geq 0$ is known as a **k-regular hypergraph**.*

Definition 6.3.5. *A hypergraph in which all edges have the same cardinality $l \geq 0$ is known as a **l-uniform hypergraph**.*

Definition 6.3.6. *The **rank** of a hypergraph $\mathcal{H}$ is defined as:*

$$r(\mathcal{H}) = \max_{D \in \mathcal{D}} |D|. \tag{6.11}$$

An edge of a hypergraph which has no vertices is called an empty edge, where the cardinality of an empty edge is 0. Furthermore, a vertex of a hypergraph which is incident to no edges is an isolated vertex, where the degree of an isolated vertex is trivially 0. An edge of cardinality 1 is a singleton and

a vertex of degree 1 is a pendant vertex. Thus a simple hypergraph $\mathcal{H}$ with $|D_i| = 2$, for every $D_i \in \mathcal{D}$ is a simple graph.

Definition 6.3.7. *The simple hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$ are **isomorphic** if there is a one-to-one correspondence between vertex sets such that any subset of vertices form an edge in $\mathcal{H}_1$ if and only if the corresponding subset of vertices form an edge in $\mathcal{H}_2$.*

We present short examples where we can use hypergraphs to model some interactions:
• Computer Science: 1) Vertices are computers in a network. Edges are subsets of computers of the same manufacturers, where there is one subset for each manufacturer.
2) Vertices are records in a relational data base, the edges are the subsets of records for which chosen attributes are true. A subset is defined for each attribute.
• Genetics: The vertices are species, the edges are sets of species with common hereditary property.
• Sociology: Vertices are employees working for a given company, edges are given interests of the employees.
• Healthcare:Vertices are different illnesses, edges are the groups of illnesses that are treated by the same medication, one hyperedge for each medication like aspirin.

Definition 6.3.8. *The **incidence matrix** of a hypergraph $\mathcal{H} = (V, \mathcal{D})$ is a matrix $I(\mathcal{H})$ with n rows and m columns for each of the edges. It takes the form:*

$$I_{ij}(\mathcal{H}) = \begin{cases} 1, & \text{if } i \in \mathcal{D}_j \\ 0, & \text{if } i \notin \mathcal{D}_j \end{cases} . \tag{6.12}$$

Note that if the edge set of hypergraphs is empty then the incidence matrix is only a column containing the names of vertices.

Definition 6.3.9. *The **dual** of a hypergraph $\mathcal{H}$ is a hypergraph $\mathcal{H}^*$ whose incidence matrix is its transpose:*

$$I(\mathcal{H}^*) = I^T(\mathcal{H}). \tag{6.13}$$

Definition 6.3.10. *For a hypergraph $\mathcal{H}$ we define the **bipartite representation** of $\mathcal{H}$ to be the bipartite graph:*

$$\mathcal{B}(\mathcal{H}) = (V, \mathcal{D}; E). \tag{6.14}$$

The vertex set in the bipartite graph is $V \cup \mathcal{D}$ where V is the left part, $\mathcal{D}$ is the right part and E is the edge set.

A vertex $i \in V$ is adjacent to a vertex $D \in \mathcal{D}$ in $\mathcal{B}(\mathcal{H})$ if and only if vertex $i \in V$ is incident to edge $D \in \mathcal{D}$. In this way, any bipartite graph is a bipartite representation of a hypergraph.

Definition 6.3.11. *For $0 \leq r \leq n$, the **complete r-uniform hypergraph** is the simple hypergraph $K_n^r = (V, \mathcal{D})$ such that $|V| = n$ and $\mathcal{D}(K_n^r)$ coincides with all the r-subsets of V.*

This means that a complete graph on n vertices is a complete 2-uniform hypergraph K_n^2, also denoted

as K_n. Furthermore, among all complete hypergraphs on four vertices it follows only K_4^2 is a simple graph. We now state without proof the following proposition.

Proposition 6.3.12. *All complete hypergraphs are r-uniform and $\binom{n-1}{r-1}$-regular hypergraphs.*

Definition 6.3.13. *In a hypergraph $\mathcal{H} = (V, \mathcal{D})$, a **path** corresponds to an alternating sequence*

$$\mu = v_0 D_0 v_1 D_1 v_2 ... v_{t-1} D_{t-1} v_t \tag{6.15}$$

of distinct nodes $v_0, v_1, v_2, ..., v_{t-1}$ and distinct edges $D_0, D_1, D_2, ..., D_{t-1}$ satisfying $v_i, v_{i+1} \in D_i$, $i = \{0, 1, ..., t-1\}$. This is connecting v_0 and v_t by a (v_0, v_t)-path.

Definition 6.3.14. *A **cycle** is a (v_0, v_t)-path in which $v_0 = v_t$.*

Definition 6.3.15. *A **connected hypergraph** is a hypergraph $\mathcal{H} = (V, \mathcal{D})$ in which for any pair of vertices there is a path between them.*

If $\mathcal{H}$ is not connected then it consists of two or more connected components where each of them is a connected hypergraph.

Definition 6.3.16. *A hypergraph $\mathcal{H} = (V, \mathcal{D})$ is **bipartite** if its vertex set can be split into two parts that are disjoint, V_1 and V_2 say, in such a way that each hyperedge of cardinality equal or bigger than 2 contains vertices from both V_1 and V_2.*

The consequence is that there is no hyperedge inside V_1 or fully inside V_2.

Definition 6.3.17. *A **complete r-partite hypergraph** is a r-uniform hypergraph $\mathcal{H} = (V, \mathcal{D})$ such that V can be partitioned into r non-empty parts.*

In this case, each edge contains precisely one vertex from each part, and all such subsets form $\mathcal{D}$.

Definition 6.3.18. *Two hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$ are **isomorphic**, denoted $\mathcal{H}_1 \equiv \mathcal{H}_2$, if there is a one-to-one correspondence between V_1 and V_2, and a one-to-one correspondence between $\mathcal{D}_1$ and $\mathcal{D}_2$ such that for each vertex $i \in V$ and for every edge $D \in \mathcal{D}$ we have that $x \in D \iff i' \in V'$ and the corresponding edge $D' \in \mathcal{D}_2$ the inclusion $i' \in D'$ holds.*

There are some basic hypergraph operations that can be used on hypergraphs.

Definition 6.3.19. *A **strong deletion** of a vertex i from $\mathcal{H}$ is removing all the edges that contain i and then taking $V \backslash \{i\}$.*

Definition 6.3.20. *A **weak deletion** of a vertex i from $\mathcal{H}$ is taking $V \backslash \{i\}$ and from each hyperedge of $\mathbb{D}(i)$.*

Similarly, edges can be strongly and weakly deleted using the same definitions as above. Now, by using strong and weak deletions we can create subhypergraphs.

Definition 6.3.21. *Assume we have a hypergraph $\mathcal{H} = (V, \mathcal{D})$. A **subhypergraph** is a hypergraph $\mathcal{H}' = (V', \mathcal{D}')$ such that $V' \subseteq V$ and $\mathcal{D}' \subseteq \mathcal{D}$.*

Note that $\mathcal{H}'$ can be obtained from $\mathcal{H}$ by strong deletion.

Definition 6.3.22. *A hypergraph $\mathcal{H}' = (V', \mathcal{D}')$ is an **induced subhypergraph** of a hypergraph $\mathcal{H} = (V, \mathcal{D})$ if $V' \subseteq V$ and all edges of $\mathcal{H}$ completely contained in V' form the family $\mathcal{D}'$. Subhypergraph $\mathcal{H}'$ may be obtained by deleting $V \backslash V'$.*

Definition 6.3.23. *For a hypergraph $\mathcal{H} = (V, \mathcal{D})$ any subhypergraph $\mathcal{H}' \subseteq \mathcal{H}$ such that $\mathcal{H}' = (V, \mathcal{D}')$ is called a **partial subhypergraph**. Such subhypergraphs have the same vertex set as the hypergraph itself and may be obtained only by weak deletion.*

Definition 6.3.24. *For a hypergraph $\mathcal{H} = (V, \mathcal{D})$ a subset of vertices $S \subset V$ is a **strongly independent set** if $|S \cap D| \leq 1$ for every hyperedge D.*

Definition 6.3.25. *A set $T \subset V$ is a **transversal** of a hypergraph $\mathcal{H} = (V, \mathcal{D})$ if $|T \cap D| \geq 1$ for every D. The cardinality of a minimum transversal is denoted by $\tau(\mathcal{H})$.*